

BITS PILANI · WILP · AIMLC ZC418

Introduction to Statistical Methods

Mid-Semester Practice Set

59 exam-style questions across 8 units, each with a full step-by-step solution right below it, a “how to attempt” plan, and a one-page cheat sheet at the end.

Descriptive statistics

Probability

Total probability & Bayes

Naïve Bayes

Discrete & continuous random variables

Joint distributions

Binomial · Poisson · Normal

Sampling & confidence intervals

Closed-book mid-semester pattern · 6 questions × 5 marks · 2 hours · standard normal table provided in the exam

Covers Contact Sessions 1 – 7

How to use this practice set

1 · Attempt before you read

Cover the solution, write your answer on paper with the same headings the solution uses (given → formula → substitution → answer). Then compare line by line.

2 · Time yourself

A 5-mark exam question gets about 18 minutes. Most questions here are 5 marks; the shorter ones are 2–3 marks. Train to finish inside that budget.

3 · Read the unit opener first

Every unit starts with what the unit is about, how to attempt its questions, and a formula card. Read those two pages before touching the questions.

4 · Learn from “where marks are lost”

Each solution ends with the mistakes that cost marks most often. Tick them off as you go — they are the difference between 3/5 and 5/5.

Colour code used in every question

Navy box — the question, with marks per part

Blue strip — what the question is testing

Gold strip — how to attempt (the plan before the working)

Green strip — answers at a glance (check your final values fast)

STEP badges — the working, one computation per line, tagged with the part it answers

Grey strip — where marks are lost

Exam-day habits that earn marks

- Write the formula in symbols *before* substituting numbers. A correct formula with an arithmetic slip still earns method marks; a bare number does not.
- Define your events or random variable in one line (“Let D = the item is defective”). Examiners look for it.
- Keep 4 decimals in probabilities and z-values. Round only at the end.
- State the method for quartiles (position $\frac{n+1}{4}$) — alternative accepted methods exist, so naming yours protects you.
- When a part says “interpret” or “comment”, answer in two sentences that quote your numbers (e.g. “mean 40.5 > median 36, so the data is positively skewed; 90 lies above the upper fence 54.5, so it is an outlier”). Vague sentences earn nothing.
- For any “at least / at most / between” probability, write the inequality first, then convert it to a sum or a complement.

Contents

Unit 1	Descriptive Statistics — central tendency, spread, five-number summary, outliers, skewness	Q 1.1 – 1.7
Unit 2	Probability Basics — axioms, addition rule, independence vs mutual exclusion, inclusion–exclusion, counting	Q 2.1 – 2.8
Unit 3	Conditional Probability, Total Probability & Bayes' Theorem	Q 3.1 – 3.8
Unit 4	Naïve Bayes Classification — tables, multiple features, text with Laplace smoothing	Q 4.1 – 4.5
Unit 5	Discrete Random Variables — pmf, expectation, variance, CDF, joint pmf, marginals, independence	Q 5.1 – 5.8
Unit 6	Continuous Random Variables — pdf, CDF, expectation, variance, joint pdf	Q 6.1 – 6.6
Unit 7	Standard Distributions — Bernoulli, Binomial, Poisson, Normal, and the approximations between them	Q 7.1 – 7.11
Unit 8	Sampling Distributions & Estimation — CLT, standard error, sample proportion, confidence intervals, sample size	Q 8.1 – 8.6
Cheat sheet	Every formula and decision rule you need, on two pages, plus the z-values you will use most	end

How the marks are usually spread

A typical paper gives one question to descriptive statistics, one to basic/conditional probability, one to Bayes or total probability, one to Naïve Bayes, one to random variables (discrete or continuous, often a joint distribution), and one to the standard distributions (Binomial/Poisson/Normal, sometimes with a sampling-distribution part). Questions are frequently split into two halves from different units (for example a 3-mark five-number summary plus a 2-mark total-probability calculation). So: **do not skip a unit** — every unit appears, and the shorter units (Unit 8, Unit 2) show up as 2-mark halves.

Notation

$\bar{x}$ sample mean · s sample standard deviation (divide by $n - 1$) · σ population standard deviation · $P(A | B)$ probability of A given B · A^c or $\bar{A}$ complement of A · $\Phi(z) = P(Z \leq z)$ from the standard normal table · $\binom{n}{k} = {}^nC_k$.

UNIT 1

Descriptive Statistics

Mean, median, mode · range, variance, standard deviation · quartiles, IQR, five-number summary · outliers (1.5 × IQR rule) · skewness · coefficient of variation

What this unit is about

This unit is about **summarising a list of numbers with a few honest numbers**. Two questions are always asked of any data set: *where is the centre?* (mean, median, mode) and *how spread out is it?* (range, variance, standard deviation, IQR). The five-number summary (Min, Q1, Median, Q3, Max) answers both at once and is the basis of the box plot and of outlier detection.

The exam question from this unit is almost always a small data set (10–20 values) followed by: sort → quartiles → IQR → fences → outliers → mean and standard deviation → a two-sentence interpretation (skewness, consistency). Occasionally you are given a ready-made summary table (count, mean, SD, quartiles) and asked to draw conclusions from it — the tools are the same: coefficient of variation, mean-vs-median for skewness, and the fences for outliers.

How to attempt questions from this unit

1. **Sort the data first** and write n . Number the positions 1 to n — almost every error in this unit comes from mis-counting positions.
2. **Quartiles by position:** Q1 at position $\frac{n+1}{4}$, Median at $\frac{n+1}{2}$, Q3 at $\frac{3(n+1)}{4}$. If the position is a fraction, interpolate between the two neighbouring values (position 2.75 means value at 2 + 0.75 × (value at 3 – value at 2)). State the method you use.
3. **IQR = Q3 – Q1**, lower fence = Q1 – 1.5 IQR, upper fence = Q3 + 1.5 IQR. A value strictly outside the fences is a potential outlier; beyond 3 × IQR it is a major outlier.
4. **Mean and SD:** compute $\bar{x}$, then a table of $x - \bar{x}$ and $(x - \bar{x})^2$, sum of squares SS, then $s^2 = SS/(n - 1)$ and $s = \sqrt{s^2}$. Use $n - 1$ for a sample (the exam data is always a sample).
5. **Interpret with numbers:** mean > median → positively (right) skewed; mean < median → negatively (left) skewed; mean ≈ median → roughly symmetric. Lower CV = more consistent. Name any outlier and say what it does to the mean.

Formula card

$$\text{Sample mean } \bar{x} = \frac{\sum x_i}{n}$$

$$\text{Sample variance } s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} = \frac{SS}{n - 1}$$

$$\text{Range} = \text{Max} - \text{Min}$$

$$\text{Quartile positions } \frac{n+1}{4}, \frac{n+1}{2}, \frac{3(n+1)}{4}$$

$$\text{Fences: } Q1 - 1.5 \text{ IQR and } Q3 + 1.5 \text{ IQR}$$

$$\text{Empirical: Mode} \approx 3 \text{ Median} - 2 \text{ Mean}$$

$$\text{Grouped mean } \bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

$$\text{Population variance } \sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

$$\text{Coefficient of variation CV} = \frac{s}{\bar{x}} \times 100\%$$

$$\text{IQR} = Q3 - Q1, \text{ Quartile deviation} = \text{IQR}/2$$

$$\text{Major outliers: beyond } Q1 - 3 \text{ IQR or } Q3 + 3 \text{ IQR}$$

$$\text{Skew: mean} > \text{median} \Rightarrow \text{right skew; mean} < \text{median} \Rightarrow \text{left skew}$$

Question map

Q	Shape of the question	Marks
1.1	Sort, five-number summary, IQR, outliers, mean, skewness ($n = 11$)	5
1.2	Mean, sample SD, CV from a deviation table, plus population SD	3
1.3	Conclusions from a ready-made summary table (CV, skew, fences)	5
1.4	Five-number summary ($n = 15$), effect of removing an outlier, box plot description	5
1.5	Reverse question: quartiles given, decide which values are outliers / major outliers	3
1.6	Even n ($n = 10$): interpolated quartiles, and the alternative “halves” method	4
1.7	Quick objective checks: mode from the empirical relation, variance from SS, scales of measurement	4

Q 1.1 Five-number summary · IQR · outliers · mean · skewness

5 marks

A courier company records the delivery time (minutes) of 11 randomly chosen parcels in a city zone:

32, 41, 28, 35, 39, 44, 30, 37, 90, 33, 36

- Find the five-number summary (Min, Q1, Median, Q3, Max) and the IQR. [2]
- Using the $1.5 \times \text{IQR}$ rule, identify any potential outliers. [1]
- Compute the mean. State whether the data is symmetric, positively skewed or negatively skewed, with reason. [1]
- Which measure of centre should the company report as the “typical” delivery time, and why? [1]

WHAT THIS QUESTION TESTS

Position-based quartiles on odd n , the fence rule, and whether you can connect mean-vs-median to skewness and to the choice of a robust centre.

HOW TO ATTEMPT

- Sort; $n = 11$, so $n + 1 = 12$ and the quartile positions are 3, 6, 9 — all whole numbers, no interpolation needed.
- IQR $\rightarrow$ fences $\rightarrow$ compare Min and Max with the fences.
- Mean = sum/11; compare with the median.
- Outlier present $\Rightarrow$ median is the better “typical” value.

ANSWERS AT A GLANCE

- Min 28, Q1 32, Median 36, Q3 41, Max 90; IQR = 9
- Fences 18.5 and 54.5 $\Rightarrow$ 90 is a potential outlier
- Mean = 40.45 > median 36 $\Rightarrow$ positively skewed
- Median (36 min) — not pulled by the 90-minute outlier

STEP-BY-STEP SOLUTION

STEP **part a** Sort and locate positions

Sorted: 28, 30, 32, 33, 35, 36, 37, 39, 41, 44, 90 ($n = 11$)

Q_1 position = $\frac{11+1}{4} = 3 \rightarrow$ 3rd value = **32**

Median position = $\frac{12}{2} = 6 \rightarrow$ 6th value = **36**

Q_3 position = $\frac{3 \times 12}{4} = 9 \rightarrow$ 9th value = **41**

Min = 28, Max = 90

$IQR = Q_3 - Q_1 = 41 - 32 = 9$

Five-number summary: 28, 32, 36, 41, 90 · IQR = 9

STEP **part b** Fences and outliers

Lower fence = $Q_1 - 1.5 IQR = 32 - 13.5 = 18.5$

Upper fence = $Q_3 + 1.5 IQR = 41 + 13.5 = 54.5$

Min 28 > 18.5 ✓ ; Max 90 > 54.5 ✗

90 minutes is a potential outlier (it even exceeds the major-outlier limit $41 + 3 \times 9 = 68$). No low outliers.

STEP **part c** Mean and skewness

Sum = $28 + 30 + 32 + 33 + 35 + 36 + 37 + 39 + 41 + 44 + 90 = 445$

$\bar{x} = 445/11 = 40.45$

Mean 40.45 > Median 36

Positively (right) skewed — the long tail is on the high side, caused by the 90-minute delivery.

STEP **part d** Choice of centre

Report the median, 36 minutes. The mean (40.45) is inflated by a single outlier; the median is unaffected by extreme values and represents a typical parcel better.

WHERE MARKS ARE LOST

Not sorting before picking quartiles · using positions $n/4$ instead of $(n+1)/4$ without saying so · comparing 90 with the mean instead of with the upper fence · writing “skewed” without the direction or the reason (mean > median).

Q 1.2 Mean · sample standard deviation · coefficient of variation**3 marks**

Daily sales (₹ thousand) of a bakery over 8 days are:

12, 15, 9, 18, 14, 11, 16, 13

- Compute the mean and the sample standard deviation. Show the deviation table. [2]
- Compute the coefficient of variation and state, in one line, what it tells the owner. [0.5]
- If these 8 days were the *entire* population of interest, what would the standard deviation be? [0.5]

WHAT THIS QUESTION TESTS

The mechanical SD computation with $n - 1$, the CV as a scale-free measure of consistency, and the difference between sample and population formulas.

HOW TO ATTEMPT

1. Mean first (it is exactly 13.5 — clean deviations).
2. Table: x , $x - \bar{x}$, $(x - \bar{x})^2$; sum of squares.
3. Divide by $n - 1 = 7$ for the sample variance, square-root.
4. CV = $s/\text{mean} \times 100$. For the population version divide SS by $n = 8$ instead.

ANSWERS AT A GLANCE

- a) Mean 13.5; SS = 58; $s^2 = 8.286$; $s = 2.878$
 b) CV $\approx 21.3\%$
 c) Population SD $\sigma = \sqrt{58/8} = 2.693$

STEP-BY-STEP SOLUTION**STEP** **part a** **Mean**

$$\text{Sum} = 12 + 15 + 9 + 18 + 14 + 11 + 16 + 13 = 108$$

$$\bar{x} = 108/8 = 13.5$$

STEP **part a** **Deviation table**

x	$x - \bar{x}$	$(x - \bar{x})^2$
12	-1.5	2.25
15	1.5	2.25
9	-4.5	20.25
18	4.5	20.25
14	0.5	0.25
11	-2.5	6.25
16	2.5	6.25
13	-0.5	0.25
Total	0	58.00

Check: deviations sum to 0 ✓ (they always must).

$$s^2 = \frac{\text{SS}}{n-1} = \frac{58}{7} = 8.2857$$

$$s = \sqrt{8.2857} = 2.878$$

Mean = ₹13.5 thousand, sample SD = ₹2.878 thousand

STEP **part b** **Coefficient of variation**

$$\text{CV} = \frac{s}{\bar{x}} \times 100 = \frac{2.878}{13.5} \times 100 = 21.3\%$$

CV $\approx 21.3\%$ — a typical day's sales deviate from the average by about a fifth of the average; useful for comparing consistency with another branch that has a different average.

STEP **part c** **Population SD**

$$\sigma = \sqrt{\frac{58}{8}} = \sqrt{7.25} = 2.693$$

Population SD = 2.693 (smaller than the sample SD because we divide by 8, not 7)

WHERE MARKS ARE LOST

Dividing by n for a sample · forgetting to square-root (reporting the variance as the SD) · arithmetic slips in the squares (write them all, one per line) · CV reported without the % sign or with s and mean swapped.

Q 1.3 Inference from a summary table · CV · skewness · fences

5 marks

Three product lines were monitored for weekly output (units). The statistical summary is:

	Line A	Line B	Line C
Count	52	52	52
Mean	120	200	60
S.D.	6	20	9
25 % (Q1)	110	180	50
50 % (Median)	118	200	64
75 % (Q3)	128	220	68
Minimum	80	150	30
Maximum	140	250	75

- a) Which line is the most consistent? Justify with the coefficient of variation. [1.5]
- b) Classify the shape of each line's distribution (symmetric / positively skewed / negatively skewed). [1.5]
- c) For each line, compute the outlier fences and state whether the minimum or maximum is a potential outlier. [2]

WHAT THIS QUESTION TESTS

Reading a summary table and drawing three *numerically justified* conclusions — consistency (CV), shape (mean vs median) and extremes (fences). No raw data is needed.

HOW TO ATTEMPT

1. $CV = SD/mean \times 100$ for each line; smallest CV wins.
2. Compare mean and median line by line.
3. $IQR = Q3 - Q1$, fences, then test Min and Max against them.

ANSWERS AT A GLANCE

- a) CV: A 5 %, B 10 %, C 15 % $\Rightarrow$ Line A most consistent
- b) A positively skewed ($120 > 118$), B symmetric ($200 = 200$), C negatively skewed ($60 < 64$)
- c) A: fences 83, 155 $\Rightarrow$ Min 80 is an outlier; B: 120, 280 $\Rightarrow$ none; C: 23, 95 $\Rightarrow$ none

STEP-BY-STEP SOLUTION**STEP** part a Coefficient of variation

A: $CV = 6/120 \times 100 = 5\%$
 B: $CV = 20/200 \times 100 = 10\%$
 C: $CV = 9/60 \times 100 = 15\%$

Line A is the most consistent (only 5 % variation about its mean). Note that B has the largest SD but is not the least consistent — SD must be compared relative to the mean.

STEP part b Shape from mean vs median

A: mean 120 > median 118 $\Rightarrow$ positively skewed (mild)
 B: mean 200 = median 200 $\Rightarrow$ symmetric
 C: mean 60 < median 64 $\Rightarrow$ negatively skewed

A: positive skew · B: symmetric · C: negative skew

STEP **part c** **Fences**

A: $IQR = 128 - 110 = 18$; fences $110 - 27 = 83$ and $128 + 27 = 155$; $\text{Min } 80 < 83 \Rightarrow \text{outlier}$; $\text{Max } 140 < 155 \Rightarrow \text{not}$
 B: $IQR = 220 - 180 = 40$; fences $180 - 60 = 120$ and $220 + 60 = 280$; $\text{Min } 150, \text{Max } 250$ both inside $\Rightarrow$ no outliers
 C: $IQR = 68 - 50 = 18$; fences $50 - 27 = 23$ and $68 + 27 = 95$; $\text{Min } 30, \text{Max } 75$ both inside $\Rightarrow$ no outliers

Only Line A has a potential outlier — its minimum week (80 units), which suggests one unusually bad week worth investigating.

WHERE MARKS ARE LOST

Saying “B varies most because $SD = 20$ ” (must use CV) · calling A “symmetric” because 120 and 118 are close — the exam wants the direction · computing fences from $\text{mean} \pm 1.5 SD$ instead of the quartiles.

Q 1.4 Five-number summary ($n = 15$) · effect of an outlier · box plot**5 marks**

Marks (out of 100) of 15 students in a quiz:

42, 55, 61, 47, 58, 63, 50, 66, 59, 71, 53, 64, 48, 60, 95

- a) Find the five-number summary and the IQR. [2]
- b) Identify outliers using the $1.5 \times IQR$ rule. [1]
- c) Compute the mean with and without the outlier. By how much does the outlier shift the mean? Does it shift the median? [1.5]
- d) Describe the box-and-whisker plot: where the box sits, where the whiskers end, and where any outlier is marked. [0.5]

WHAT THIS QUESTION TESTS

Quartiles with $n = 15$ (positions 4, 8, 12), and the key property that the mean is sensitive to outliers while the median and quartiles are not.

HOW TO ATTEMPT

- Sort; positions 4, 8, 12.
- Fences; 95 is the suspect.
- Sum of all 15, then subtract 95 and divide by 14.
- Whiskers stop at the last data value *inside* the fences, not at the fence itself.

ANSWERS AT A GLANCE

- a) 42, 50, 59, 64, 95; $IQR = 14$
- b) Fences 29 and 85 $\Rightarrow$ 95 is an outlier
- c) Mean 59.47 with, 56.93 without (shift ≈ 2.5 marks); median 59 $\rightarrow$ 58.5 (barely moves)
- d) Box from 50 to 64 with the median line at 59; whiskers to 42 and 71; 95 plotted as a separate point

STEP-BY-STEP SOLUTION**STEP** **part a** **Sort and pick positions**

Sorted: 42, 47, 48, 50, 53, 55, 58, 59, 60, 61, 63, 64, 66, 71, 95 ($n = 15$, $n + 1 = 16$)

Q1 at position $16/4 = 4 \rightarrow 50$; Median at 8 $\rightarrow 59$; Q3 at 12 $\rightarrow 64$

$IQR = 64 - 50 = 14$

Min 42, Q1 50, Median 59, Q3 64, Max 95; $IQR = 14$

STEP **part b** **Fences**

$$\text{Lower fence} = 50 - 1.5 \times 14 = 50 - 21 = 29$$

$$\text{Upper fence} = 64 + 21 = 85$$

$95 > 85 \Rightarrow 95$ is a potential outlier; $42 > 29 \Rightarrow$ no low outlier.

STEP **part c** **Mean with and without the outlier**

$$\text{Sum of all 15} = 892, \text{ so } \bar{x} = 892/15 = 59.47$$

$$\text{Without 95: sum} = 797, n = 14, \bar{x} = 797/14 = 56.93$$

$$\text{Shift} = 59.47 - 56.93 = 2.54 \text{ marks}$$

$$\text{Median without 95: 14 values, positions 7 and 8} \rightarrow (58 + 59)/2 = 58.5$$

The single outlier pulls the mean up by about 2.5 marks; the median moves only from 59 to 58.5. This is why the median is preferred for skewed data.

STEP **part d** **Box plot description**

Box from $Q1 = 50$ to $Q3 = 64$, median line at 59 (slightly right of centre $\rightarrow$ mild left skew inside the box). Lower whisker to 42 (smallest value inside the fences), upper whisker to 71 (largest value inside 85). The value 95 is drawn as an isolated point beyond the upper whisker.

WHERE MARKS ARE LOST

Drawing the upper whisker to 95 (an outlier is never the whisker end) · recomputing the median with 15 values after removing one · forgetting that n changes to 14 in part (c).

Q 1.5 Outlier rule in reverse · major vs potential outliers**3 marks**

For a large data set you are told only that $Q1 = 20$ and $Q3 = 32$.

- a)** Which of the following observations would be flagged as potential outliers: 0, 2, 18, 50, 52? [1.5]
- b)** What is the smallest value that would be flagged as a *major* outlier on the high side? [0.5]
- c)** The data set has 200 values and a box plot shows both whiskers ending exactly at the fences. [1]
What is the maximum possible range of the non-outlier data?

WHAT THIS QUESTION TESTS

Whether you can apply the fences without data in hand, and whether you know a value *on* the fence is not an outlier.

HOW TO ATTEMPT

- $IQR = 12$; fences 2 and 50; major fences -16 and 68.
- Outlier means strictly outside the fence.
- Non-outlier range $\leq$ upper fence - lower fence.

ANSWERS AT A GLANCE

- a) $IQR = 12$, fences 2 and 50 $\Rightarrow$ outliers: 0 and 52 (2 and 50 sit on the fence; 18 is inside)
- b) $Q3 + 3 IQR = 68 \Rightarrow$ any value above 68 (the smallest such observation would be just above 68)
- c) $50 - 2 = 48$

STEP-BY-STEP SOLUTION

STEP **part a** **Fences**

$$\text{IQR} = 32 - 20 = 12; 1.5 \times \text{IQR} = 18$$

$$\text{Lower fence} = 20 - 18 = 2; \text{upper fence} = 32 + 18 = 50$$

$$0 < 2 \Rightarrow \text{outlier}; 2 = \text{fence} \Rightarrow \text{not an outlier}; 18 \text{ inside}; 50 = \text{fence} \Rightarrow \text{not}; 52 > 50 \Rightarrow \text{outlier}$$

Potential outliers: 0 and 52

STEP **part b** **Major outlier limit**

$$Q_3 + 3 \times \text{IQR} = 32 + 36 = 68$$

Values above 68 are major outliers (and the lower major limit is $20 - 36 = -16$).

STEP **part c** **Range of the retained data**

All non-outliers lie within $[2, 50]$

Maximum possible range = $50 - 2 = 48$

WHERE MARKS ARE LOST

Treating a value equal to the fence as an outlier · using $1.5 \times \text{IQR}$ for the major-outlier limit · confusing “range of the whole data” with the range after outliers are removed.

Q 1.6 Even n · interpolated quartiles · two accepted methods

4 marks

Response times (seconds) of a help-desk chatbot for 10 queries:

12, 15, 11, 19, 14, 22, 13, 17, 16, 40

- a) Find the five-number summary using the $\frac{n+1}{4}$ position method with interpolation. [2]
- b) Identify outliers. [1]
- c) Repeat Q1 and Q3 by the “median of the lower half / upper half” method. Does the outlier decision change? [1]

WHAT THIS QUESTION TESTS

Handling a fractional quartile position, and the reassurance that the two common methods give slightly different quartiles but (almost always) the same conclusion.

HOW TO ATTEMPT

- Sort; $n = 10$, positions 2.75, 5.5, 8.25.
- Interpolate: value at $2 + 0.75 \times (\text{value at } 3 - \text{value at } 2)$, etc.
- Halves method: lower half = first 5 values, upper half = last 5 values; take their medians.

ANSWERS AT A GLANCE

- a) Min 11, Q1 12.75, Median 15.5, Q3 19.75, Max 40; IQR 7
- b) Fences 2.25 and 30.25 $\Rightarrow$ 40 is an outlier
- c) Halves method: Q1 = 13, Q3 = 19, IQR 6, fences 4 and 28 $\Rightarrow$ 40 still an outlier

STEP-BY-STEP SOLUTION

STEP **part a** Interpolated positions

Sorted: 11, 12, 13, 14, 15, 16, 17, 19, 22, 40 ($n = 10$, $n + 1 = 11$)

Q1 position = $11/4 = 2.75 \rightarrow 12 + 0.75(13 - 12) = 12.75$

Median position = $5.5 \rightarrow (15 + 16)/2 = 15.5$

Q3 position = $8.25 \rightarrow 19 + 0.25(22 - 19) = 19.75$

IQR = $19.75 - 12.75 = 7$

11, 12.75, 15.5, 19.75, 40; IQR = 7

STEP **part b** Outliers

Fences: $12.75 - 10.5 = 2.25$ and $19.75 + 10.5 = 30.25$

$40 > 30.25 \Rightarrow 40$ seconds is a potential outlier.

STEP **part c** Halves method

Lower half 11, 12, 13, 14, 15 $\rightarrow$ median 13 = Q1

Upper half 16, 17, 19, 22, 40 $\rightarrow$ median 19 = Q3

IQR = 6; fences $13 - 9 = 4$ and $19 + 9 = 28$

$40 > 28 \Rightarrow$ same decision: 40 is an outlier. Quartiles differ slightly by method; state which one you use.

WHERE MARKS ARE LOST

Interpolating in the wrong direction (position 2.75 is closer to the 3rd value, so add 0.75 of the gap to the 2nd value) · mixing the two methods in one answer · forgetting the median is the average of the 5th and 6th values when n is even.

Q 1.7 Quick objective checks · empirical relation · SS · measurement scales

4 marks

- a) A moderately skewed distribution has mean 50 and median 54. Estimate the mode using the empirical relation and state the direction of skew. [1]
- b) A sample of 5 scores has sum of squared deviations $SS = 48$. Find the sample variance and standard deviation. [1]
- c) Classify each variable by level of measurement (nominal / ordinal / interval / ratio) and, where numerical, as discrete or continuous: (i) blood group, (ii) customer rating 1–5 stars, (iii) temperature in $^{\circ}\text{C}$, (iv) number of app crashes per day, (v) monthly rent in ₹. [1.5]
- d) True or false, with a one-line reason: "The standard deviation is the distance from the mean to the farthest observation." [0.5]

WHAT THIS QUESTION TESTS

Short recall items that appear as half-questions: the empirical relation $\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$, variance from SS, and measurement scales.

HOW TO ATTEMPT

1. Plug into $\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$.
2. $s^2 = SS/(n - 1)$.
3. Nominal = names only; ordinal = order but no equal gaps; interval = equal gaps, no true zero; ratio = true zero.

ANSWERS AT A GLANCE

- a) Mode ≈ 62 ; mean $<$ median $<$ mode $\Rightarrow$ negatively skewed
 b) $s^2 = 12$, $s = 3.464$
 c) (i) nominal (ii) ordinal (iii) interval, continuous (iv) ratio, discrete (v) ratio, continuous
 d) False — SD is a typical (root-mean-square) distance from the mean, not the maximum distance

STEP-BY-STEP SOLUTION**STEP** **part a** Empirical relation

$$\text{Mode} \approx 3(54) - 2(50) = 162 - 100 = 62$$

Mode ≈ 62 . Since mean (50) $<$ median (54) $<$ mode (62), the distribution is negatively (left) skewed.

STEP **part b** Variance from SS

$$s^2 = \frac{48}{5 - 1} = 12, \quad s = \sqrt{12} = 3.464$$

Variance 12, SD 3.464

STEP **part c** Scales

- (i) Blood group — categories with no order $\Rightarrow$ nominal
 (ii) Star rating — ordered categories, gaps not necessarily equal $\Rightarrow$ ordinal
 (iii) °C — equal gaps, 0 °C is not “no temperature” $\Rightarrow$ interval; continuous
 (iv) Crashes per day — counts with a true zero $\Rightarrow$ ratio; discrete
 (v) Rent — true zero, ratios meaningful $\Rightarrow$ ratio; continuous

STEP **part d** True/false

False. SD = $\sqrt{\text{average squared deviation}}$; it summarises the spread of *all* values, and is always smaller than the largest deviation.

WHERE MARKS ARE LOST

Writing Mode = 3 Mean – 2 Median (reversed) · dividing SS by n · labelling temperature in °C as ratio.

UNIT 2

Probability Basics

Sample space and events · axioms · addition rule · mutually exclusive vs independent · complements · inclusion–exclusion for two and three events · counting with combinations

What this unit is about

Probability is a **bookkeeping system for uncertainty**: every outcome gets a non-negative number, the numbers for the whole sample space add to 1, and the probability of “A or B” is found by adding and then removing the overlap you counted twice. That last idea — *inclusion–exclusion* — is the workhorse of this unit and appears in every paper in some form (two events, or three events with “exactly one / exactly two / at least one / none”).

Two words are confused constantly and the exam deliberately tests the difference. **Mutually exclusive** means the events cannot happen together: $P(A \cap B) = 0$. **Independent** means one happening tells you nothing about the other: $P(A \cap B) = P(A)P(B)$. Two events with positive probabilities cannot be both — if they are mutually exclusive, knowing A happened tells you B did not, so they are dependent.

How to attempt questions from this unit

1. **Name the events** in one line each and write every given number in symbols ($P(A) = 0.3$, $P(B | A) = 0.6$...). Half the marks are in translating the words correctly: “of those who ...” is a conditional; “both” is an intersection; “or / at least one” is a union; “neither” is the complement of the union.
2. **Two events**: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Exactly one = $P(A) + P(B) - 2P(A \cap B)$. Neither = $1 - P(A \cup B)$. Only A = $P(A) - P(A \cap B)$.
3. **Three events (counts or probabilities)**: write the three-circle Venn diagram from the inside out — start with all three, then each pair minus the triple, then each single minus its pairs plus the triple. Every “exactly / only / at least” question becomes adding the right regions.
4. **Test independence** by comparing $P(A \cap B)$ with $P(A)P(B)$ (or $P(A | B)$ with $P(A)$). Say “equal $\Rightarrow$ independent” or “not equal $\Rightarrow$ dependent”.
5. **Counting**: equally likely selections use $\binom{n}{k}$; “at least one” is almost always $1 - P(\text{none})$; with replacement multiplies the same probability, without replacement changes the denominator each draw.

Formula card

Axioms: $0 \leq P(A) \leq 1$, $P(S) = 1$; disjoint $\Rightarrow P(A \cup B) = P(A) + P(B)$

Addition: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Mutually exclusive: $P(A \cap B) = 0$

Three events: $P(A \cup B \cup C) = \sum P(A) - \sum P(A \cap B) + P(A \cap B \cap C)$

None: $1 - P(A \cup B \cup C)$

De Morgan: $P(A^c \cap B^c) = 1 - P(A \cup B)$; $P(A^c \cup B^c) = 1 - P(A \cap B)$

Complement: $P(A^c) = 1 - P(A)$

Exactly one of A, B: $P(A) + P(B) - 2P(A \cap B)$

Independent: $P(A \cap B) = P(A)P(B)$, and then A^c, B^c are independent too

Exactly two of three: $\sum P(\text{pairs}) - 3P(A \cap B \cap C)$

Only A: $P(A) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$

Combinations: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Question map

Q	Shape of the question	Marks
2.1	Permissible probability assignments; complements and unions of disjoint events	3
2.2	Independent events: validate statements, compute intersections, unions, complements, conditionals	5
2.3	Two dependent events given $P(A)$, $P(B A)$, $P(A B)$: recover $P(B)$ and everything else	5
2.4	Three sets with counts: all three, exactly two, only one, "P or S but not C"	6
2.5	Three symmetric events: recover $P(A)$ from $P(\text{none})$, then exactly one / two, at least one	5
2.6	Two-way count table: conditional probabilities and an independence check	4
2.7	Selection without / with replacement using combinations	4
2.8	Contradiction between two witnesses; "at least one success" and how many trials are needed	4

Q 2.1 Axioms · permissible assignments · disjoint events

3 marks

An experiment has four mutually exclusive and exhaustive outcomes A, B, C, D.

- a) Decide whether each assignment is permissible, giving the reason: (i) 0.36, 0.24, 0.18, 0.22 (ii) 0.40, 0.30, 0.25, 0.15 (iii) 0.50, 0.30, -0.05, 0.25 (iv) $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$. [2]
- b) Using assignment (iv), find $P(A \cup C)$, $P(B^c)$ and $P(A^c \cap D^c)$. [1]

WHAT THIS QUESTION TESTS

The two axioms (non-negative, total 1) and the fact that disjoint events simply add.

HOW TO ATTEMPT

- Check every number is in $[0, 1]$ and the four add to exactly 1.
- For disjoint outcomes, union = sum. $A^c \cap D^c = \text{"neither A nor D"} = B \text{ or } C$.

ANSWERS AT A GLANCE

- a) (i) yes (sum 1) (ii) no (sum 1.10) (iii) no (negative) (iv) yes (sum 1)
 b) $P(A \cup C) = 5/8$, $P(B^c) = 3/4$, $P(A^c \cap D^c) = 3/8$

STEP-BY-STEP SOLUTION

STEP part a Check the two axioms

- (i) all in $[0, 1]$; $0.36 + 0.24 + 0.18 + 0.22 = 1.00 \Rightarrow$ permissible
 (ii) $0.40 + 0.30 + 0.25 + 0.15 = 1.10 \neq 1 \Rightarrow$ not permissible
 (iii) $P(C) = -0.05 < 0 \Rightarrow$ not permissible (sum is irrelevant once a value is negative)
 (iv) $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = 1 \Rightarrow$ permissible

STEP part b Disjoint events add

$$P(A \cup C) = \frac{1}{2} + \frac{1}{8} = \frac{5}{8}$$

$$P(B^c) = 1 - \frac{1}{4} = \frac{3}{4}$$

$$P(A^c \cap D^c) = P(B \cup C) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8} \text{ (equivalently } 1 - P(A \cup D) = 1 - \frac{5}{8} \text{)}$$

5/8, 3/4, 3/8

WHERE MARKS ARE LOST

Accepting (ii) because "each value is between 0 and 1" · subtracting $P(A \cap C)$ that is zero anyway but forgetting to say why (disjoint) · reading $A^c \cap D^c$ as "not A or not D".

Q 2.2 Independent events · validate statements · complements · conditionals

5 marks

Two independent events A and B have $P(A) = 0.40$ and $P(B) = 0.25$.

- a) A student claims "since A and B are independent, $P(A \cup B) = P(A) + P(B) = 0.65$ ". Is the claim correct? Compute the correct value. [1]
- b) Another student claims "A and B are mutually exclusive, so $P(A \cap B) = 0$ ". Validate. [1]
- c) Find $P(A | B)$, $P(A^c | B)$ and $P(A^c \cap B^c)$. [2]
- d) Find the probability that exactly one of A and B occurs. [1]

WHAT THIS QUESTION TESTS

The definition of independence, and the fact that independent events with positive probability can never be mutually exclusive.

HOW TO ATTEMPT

1. Independence gives $P(A \cap B) = P(A)P(B)$ — compute it first, everything else follows.
2. Addition rule for the union.
3. For independent events $P(A | B) = P(A)$; complements stay independent.
4. Exactly one = union – intersection.

ANSWERS AT A GLANCE

- a) Wrong; $P(A \cap B) = 0.10$, $P(A \cup B) = 0.55$
- b) Wrong; $P(A \cap B) = 0.10 \neq 0$
- c) $P(A | B) = 0.40$, $P(A^c | B) = 0.60$, $P(A^c \cap B^c) = 0.45$
- d) 0.45

STEP-BY-STEP SOLUTION**STEP** part a Intersection first, then the union

$$\text{Independent} \Rightarrow P(A \cap B) = P(A)P(B) = 0.40 \times 0.25 = 0.10$$

$$P(A \cup B) = 0.40 + 0.25 - 0.10 = 0.55$$

The claim is wrong. "Add without subtracting" is only valid for mutually exclusive events. Correct value 0.55.

STEP part b Mutually exclusive?

$$P(A \cap B) = 0.10 \neq 0$$

Not mutually exclusive. Independent events with non-zero probabilities always overlap: $P(A \cap B) = P(A)P(B) > 0$.

STEP part c Conditionals and complements

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0.10}{0.25} = 0.40 = P(A) \text{ (as expected under independence)}$$

$$P(A^c | B) = 1 - P(A | B) = 0.60$$

$$P(A^c) = 0.60, P(B^c) = 0.75; \text{ complements of independent events are independent}$$

$$P(A^c \cap B^c) = 0.60 \times 0.75 = 0.45 \text{ (check: } 1 - P(A \cup B) = 1 - 0.55 = 0.45 \checkmark)$$

0.40, 0.60, 0.45

STEP part d **Exactly one**

$$P(\text{exactly one}) = P(A \cup B) - P(A \cap B) = 0.55 - 0.10 = 0.45$$

$$(\text{or } P(A)P(B^c) + P(A^c)P(B) = 0.40 \times 0.75 + 0.60 \times 0.25 = 0.30 + 0.15 = 0.45)$$

0.45**WHERE MARKS ARE LOST**

Saying “independent, so they can’t happen together” · computing $P(A^c \cap B^c)$ as $1 - P(A \cap B)$ (that is $P(A^c \cup B^c)$) · in (d) reporting the union instead of exactly one.

Q 2.3 Dependent events · $P(A)$, $P(B|A)$, $P(A|B)$ given · recover $P(B)$ **5 marks**

At an online store, 30 % of customers order through the mobile app (event A). Among app customers, 60 % pay by UPI (event U). Among all UPI payers, 45 % are app customers. A customer is chosen at random. Find the probability that the customer

- a) uses the app and pays by UPI; [1]
- b) uses the app or pays by UPI; [1]
- c) does not use the app, given that they pay by UPI; [1]
- d) neither uses the app nor pays by UPI; [1]
- e) uses the app but does not pay by UPI. Are A and U independent? [1]

WHAT THIS QUESTION TESTS

Reading three different kinds of percentages (marginal, conditional one way, conditional the other way) and using the multiplication rule twice — once to get the intersection, once backwards to get the missing marginal.

HOW TO ATTEMPT

- Write $P(A) = 0.30$, $P(U | A) = 0.60$, $P(A | U) = 0.45$.
- $P(A \cap U) = P(A)P(U | A)$.
- Then $P(U) = P(A \cap U)/P(A | U)$ — the missing marginal.
- Union, complement of the union, and “A only”; compare $P(A \cap U)$ with $P(A)P(U)$.

ANSWERS AT A GLANCE

a) 0.18 b) 0.52 c) 0.55 d) 0.48 e) 0.12; not independent ($0.18 \neq 0.12$)

STEP-BY-STEP SOLUTION**STEP** part a **Intersection by the multiplication rule**

$$\text{Given: } P(A) = 0.30, \quad P(U | A) = 0.60, \quad P(A | U) = 0.45$$

$$P(A \cap U) = P(A)P(U | A) = 0.30 \times 0.60 = 0.18$$

0.18**STEP** part b **Recover $P(U)$, then the union**

$$P(A | U) = \frac{P(A \cap U)}{P(U)} \Rightarrow P(U) = \frac{0.18}{0.45} = 0.40$$

$$P(A \cup U) = 0.30 + 0.40 - 0.18 = 0.52$$

0.52

STEP **part c** Conditional complement

$$P(A^c | U) = 1 - P(A | U) = 1 - 0.45 = 0.55$$

0.55**STEP** **part d** Neither

$$P(A^c \cap U^c) = 1 - P(A \cup U) = 1 - 0.52 = 0.48$$

0.48**STEP** **part e** A only, and independence

$$P(A \cap U^c) = P(A) - P(A \cap U) = 0.30 - 0.18 = 0.12$$

$$\text{Independence test: } P(A)P(U) = 0.30 \times 0.40 = 0.12 \neq 0.18 = P(A \cap U)$$

0.12; A and U are dependent (app users are more likely to pay by UPI: $P(U | A) = 0.60 > P(U) = 0.40$).**WHERE MARKS ARE LOST**

Treating “45 % of UPI payers are app customers” as $P(U | A)$ · forgetting to recover $P(U)$ before the union · in (e) using $P(A)P(U^c)$, which assumes independence.

Q 2.4 Three sets with counts · inclusion–exclusion · only / exactly / but-not**6 marks**

A company audited the certifications of its 400 engineers in Python (P), SQL (S) and Cloud (C): $|P| = 230$, $|S| = 190$, $|C| = 160$, $|P \cap S| = 100$, $|P \cap C| = 80$, $|S \cap C| = 70$. Every engineer holds at least one of the three certifications. An engineer is picked at random.

- a) How many engineers hold all three? Give the probability. [1]
- b) Probability of holding exactly two certifications. [1]
- c) Probability of holding Python but not Cloud. [1]
- d) Probability of holding only SQL. [1]
- e) Probability of holding Python or SQL but not Cloud. [1]
- f) Verify that your seven Venn regions add to 400. [1]

WHAT THIS QUESTION TESTS

Three-set inclusion–exclusion, and building the Venn diagram from the inside out so that every “only / exactly / but not” part becomes a sum of regions.

HOW TO ATTEMPT

- “Everyone has at least one” $\Rightarrow |P \cup S \cup C| = 400$; solve inclusion–exclusion for the triple.
- Fill the Venn regions: triple $\rightarrow$ pair-only regions $\rightarrow$ single-only regions.
- Answer each part by adding regions; divide by 400.

ANSWERS AT A GLANCE

a) 70 engineers, 0.175 b) 0.10 c) 0.375 d) 0.225 e) 0.60 f) $120+90+80+30+10+0+70 = 400 \checkmark$

STEP-BY-STEP SOLUTION

STEP **part a** Inclusion–exclusion for the triple

$$|P \cup S \cup C| = |P| + |S| + |C| - |P \cap S| - |P \cap C| - |S \cap C| + |P \cap S \cap C|$$

$$400 = 230 + 190 + 160 - 100 - 80 - 70 + x = 330 + x$$

$$x = 70; P(\text{all three}) = 70/400 = 0.175$$

70 engineers; probability 0.175

STEP **Build the Venn regions (inside out)**

$$P \cap S \text{ only} = 100 - 70 = 30; P \cap C \text{ only} = 80 - 70 = 10; S \cap C \text{ only} = 70 - 70 = 0$$

$$\text{Only } P = 230 - 100 - 80 + 70 = 120$$

$$\text{Only } S = 190 - 100 - 70 + 70 = 90$$

$$\text{Only } C = 160 - 80 - 70 + 70 = 80$$

Only P	Only S	Only C	P ∩ S only	P ∩ C only	S ∩ C only	All three
120	90	80	30	10	0	70

STEP **part b** Exactly two

$$\text{Exactly two} = 30 + 10 + 0 = 40 \text{ (formula: } (100 + 80 + 70) - 3 \times 70 = 40)$$

$$40/400 = 0.10$$

STEP **part c** Python but not Cloud

$$|P| - |P \cap C| = 230 - 80 = 150 \text{ (regions: only } P \text{ 120 + } P \cap S \text{ only 30)}$$

$$150/400 = 0.375$$

STEP **part d** Only SQL

$$90/400 = 0.225$$

STEP **part e** P or S, but not C

$$\text{Regions of } P \cup S \text{ outside } C: \text{only } P \text{ 120 + only } S \text{ 90 + } P \cap S \text{ only 30} = 240$$

$$\text{(check: } |P \cup S| = 230 + 190 - 100 = 320; \text{ part of it inside } C = 80 + 70 - 70 = 80; 320 - 80 = 240 \checkmark)$$

$$240/400 = 0.60$$

STEP **part f** Region check

$$120 + 90 + 80 + 30 + 10 + 0 + 70 = 400 \checkmark \text{ — consistent with “everyone holds at least one”}.$$

WHERE MARKS ARE LOST

Using “exactly two” = sum of the three pairwise counts (each pair still contains the triple) · “P but not C” computed as $|P| - |C|$ · forgetting to divide counts by 400 when a probability is asked.

Q 2.5 Three symmetric events · recover $P(A)$ from $P(\text{none})$ · exactly one / two

5 marks

Three independent-looking alerts A, B, C on a server dashboard are equally likely: $P(A) = P(B) = P(C) = p$. Any two alerts fire together with probability 0.10, all three with probability 0.04, and the probability that *no* alert fires on a given day is 0.36.

- Find p . [2]
- Probability that exactly two alerts fire. [1]
- Probability that at least one alert fires. [0.5]
- Probability that exactly one alert fires. [1]
- Are A and B independent? [0.5]

WHAT THIS QUESTION TESTS

Running inclusion–exclusion backwards (solve for an unknown), then decomposing the union into exactly-one, exactly-two and all-three pieces.

HOW TO ATTEMPT

1. “None” = $1 - P(A \cup B \cup C)$; expand with inclusion–exclusion and solve for p .
2. Exactly two = pairs – 3 × triple. Exactly one = at least one – exactly two – all three.
3. Independence: compare $P(A \cap B) = 0.10$ with p^2 .

ANSWERS AT A GLANCE

a) $p = 0.30$ b) 0.18 c) 0.64 d) 0.42 e) No: $0.30^2 = 0.09 \neq 0.10$

STEP-BY-STEP SOLUTION**STEP** **part a** Inclusion–exclusion backwards

$$P(\text{none}) = 1 - P(A \cup B \cup C)$$

$$P(A \cup B \cup C) = 3p - 3(0.10) + 0.04 = 3p - 0.26$$

$$0.36 = 1 - (3p - 0.26) = 1.26 - 3p \Rightarrow 3p = 0.90$$

$$p = 0.30$$

STEP **part b** Exactly two

$$P(\text{exactly two}) = [P(A \cap B) + P(B \cap C) + P(A \cap C)] - 3P(A \cap B \cap C) = 0.30 - 0.12 = 0.18$$

$$0.18$$

STEP **part c** At least one

$$P(\text{at least one}) = 1 - 0.36 = 0.64$$

$$0.64$$

STEP **part d** Exactly one

$$P(\text{exactly one}) = 0.64 - 0.18 - 0.04 = 0.42$$

(check via “only A” = $0.30 - 0.10 - 0.10 + 0.04 = 0.14$, and $3 \times 0.14 = 0.42$ ✓)

$$0.42$$

STEP **part e** Independence

$$P(A)P(B) = 0.30 \times 0.30 = 0.09 \neq 0.10 = P(A \cap B)$$

Not independent (the alerts are slightly positively associated).

WHERE MARKS ARE LOST

Forgetting the “+ triple” term in inclusion–exclusion · using pairs – triple (instead of – 3 × triple) for exactly two · assuming independence because the question said “independent-looking”.

Q 2.6 Two-way table · conditional probability · independence check

4 marks

200 orders were classified by region and by whether delivery was late:

	Late	On time	Total
North	18	102	120
South	12	68	80
Total	30	170	200

- a) $P(\text{Late})$ and $P(\text{Late} \mid \text{North})$. [1]
 b) $P(\text{North} \mid \text{Late})$. [1]
 c) Are "North" and "Late" independent? Justify two ways. [1]
 d) $P(\text{North} \cup \text{Late})$. [1]

WHAT THIS QUESTION TESTS

Conditional probability as a ratio of cells to a row or column total, and both ways of checking independence.

HOW TO ATTEMPT

- Conditioning on a row: divide by the row total. Conditioning on a column: divide by the column total.
- Independence: $P(L \mid N) = P(L)$? and $P(N \cap L) = P(N)P(L)$?

ANSWERS AT A GLANCE

a) 0.15 and 0.15 b) 0.60 c) Yes — $P(L \mid N) = P(L) = 0.15$ and $0.09 = 0.6 \times 0.15$ d) 0.66

STEP-BY-STEP SOLUTION**STEP** **part a** **Marginal and conditional**

$$P(L) = 30/200 = 0.15$$

$$P(L \mid N) = 18/120 = 0.15$$

STEP **part b** **Condition on the column**

$$P(N \mid L) = 18/30 = 0.60$$

STEP **part c** **Independence, two ways**

Way 1: $P(L \mid N) = 0.15 = P(L) \Rightarrow$ knowing the region does not change the late-rate

Way 2: $P(N \cap L) = 18/200 = 0.09$; $P(N)P(L) = 0.60 \times 0.15 = 0.09 \Rightarrow$ equal

Independent — North and South have the same 15 % late-rate.

STEP **part d** **Union**

$$P(N \cup L) = 0.60 + 0.15 - 0.09 = 0.66$$

0.66

WHERE MARKS ARE LOST

Dividing by 200 for a conditional probability · concluding "dependent because the counts 18 and 12 differ" (rates matter, not counts) · in (d) adding $0.60 + 0.15$ without removing the overlap.

Q 2.7 Selection without / with replacement · combinations

4 marks

A batch of 12 laptops contains 3 with a faulty battery. Two laptops are picked at random without replacement. Find the probability that

- a) both are fault-free; [1]
- b) exactly one is faulty; [1]
- c) at least one is faulty; [1]
- d) Repeat (a) if the first laptop is put back before the second is picked. [1]

WHAT THIS QUESTION TESTS

Equally likely selections via $\binom{n}{k}$ (or the sequential multiplication rule), the complement trick for “at least one”, and how replacement changes the answer.

HOW TO ATTEMPT

- Total ways $\binom{12}{2} = 66$.
- Favourable ways: choose from the 9 good and 3 faulty appropriately.
- At least one faulty = 1 – both good.
- With replacement: draws are independent, $(9/12)^2$.

ANSWERS AT A GLANCE

a) $36/66 = 6/11 = 0.5455$ b) $27/66 = 9/22 = 0.4091$ c) $5/11 = 0.4545$ d) 0.5625

STEP-BY-STEP SOLUTION**STEP** part a Both good

Total selections $\binom{12}{2} = 66$; good pairs $\binom{9}{2} = 36$

$P = 36/66 = 6/11 = 0.5455$ (sequentially: $\frac{9}{12} \times \frac{8}{11} = 0.5455 \checkmark$)

0.5455

STEP part b Exactly one faulty

Favourable = $\binom{9}{1}\binom{3}{1} = 27$; $P = 27/66 = 0.4091$

0.4091

STEP part c At least one faulty

$P = 1 - P(\text{both good}) = 1 - 6/11 = 5/11 = 0.4545$

(check: exactly one $27/66$ + both faulty $\binom{3}{2}/66 = 3/66$ gives $30/66 = 0.4545 \checkmark$)

0.4545

STEP part d With replacement

$P = \left(\frac{9}{12}\right)^2 = 0.75^2 = 0.5625$

0.5625 — slightly higher than 0.5455 because the second draw is no longer “helped” by having removed a good laptop.

WHERE MARKS ARE LOST

Using $9/12 \times 9/12$ without replacement · exactly one = 9×3 but forgetting the 66 denominator · “at least one” computed as $P(\text{first faulty}) + P(\text{second faulty})$.

Q 2.8 Independent trials · contradiction · at least one success · minimum number of trials

4 marks

- a) Witness A tells the truth 70 % of the time and witness B 80 % of the time, independently. In what percentage of cases will they contradict each other about the same fact? [1.5]
- b) A basketball player scores a free throw with probability 0.2 on each independent attempt. Find the probability of at least one score in 3 attempts. [1]
- c) How many attempts must the player take so that the probability of at least one score exceeds 0.90? [1.5]

WHAT THIS QUESTION TESTS

Building compound events from independent pieces ("A true and B false, or A false and B true"), the complement rule, and solving an inequality in n with logarithms.

HOW TO ATTEMPT

1. Contradict = exactly one tells the truth.
2. At least one = $1 - (\text{all miss})$.
3. Set $1 - 0.8^n > 0.9$, take logs (both logs are negative — flip the inequality).

ANSWERS AT A GLANCE

a) 0.38 = 38 % b) 0.488 c) $n \geq 11$

STEP-BY-STEP SOLUTION**STEP** **part a** **Contradiction = exactly one truthful**

$$P = P(A \text{ true})P(B \text{ false}) + P(A \text{ false})P(B \text{ true}) = 0.7(0.2) + 0.3(0.8) = 0.14 + 0.24 = 0.38$$

38 % of cases

STEP **part b** **At least one in 3**

$$P(\text{at least one}) = 1 - 0.8^3 = 1 - 0.512 = 0.488$$

0.488

STEP **part c** **Solve for n**

$$1 - 0.8^n > 0.90 \Rightarrow 0.8^n < 0.10$$

$$n \ln 0.8 < \ln 0.1 \Rightarrow n > \frac{\ln 0.1}{\ln 0.8} = \frac{-2.3026}{-0.2231} = 10.32$$

$$\text{Check: } 0.8^{10} = 0.107 \text{ (not enough), } 0.8^{11} = 0.086 \text{ (enough)}$$

At least 11 attempts

WHERE MARKS ARE LOST

Contradiction computed as 0.3×0.2 (that is both lying, which is agreement) · dividing the two logarithms without flipping the inequality sign · answering 10 instead of rounding up to 11.

UNIT 3

Conditional Probability, Total Probability & Bayes' Theorem

Multiplication rule · tree diagrams · law of total probability · Bayes' theorem with two or more causes · posterior probabilities for tests, alarms and diagnostics

What this unit is about

This unit is about **updating a probability when you learn something**. Conditional probability $P(B | A) = P(A \cap B)/P(A)$ is the definition; the multiplication rule $P(A \cap B) = P(A)P(B | A)$ is the same thing rearranged. Stack several causes and you get the **law of total probability** — the probability of an effect is a weighted average of its probability under each cause, weighted by how likely each cause is. Run that backwards and you get **Bayes' theorem** — given the effect happened, which cause is most likely?

Almost every exam question here has the same skeleton: 2–3 causes with prior probabilities (they add to 1), an effect with a conditional probability under each cause, then “what is the probability of the effect?” (total probability) and “given the effect, what is the probability of cause i ?” (Bayes). The famous special case is a diagnostic test for a rare condition: even a good test gives a surprisingly small posterior because the condition is rare. The exam loves asking you to comment on that.

How to attempt questions from this unit

1. **Draw the tree.** First branches = causes with priors; second branches = effect / no effect with conditional probabilities. Multiply along a path to get a joint probability $P(\text{cause} \cap \text{effect})$.
2. **Total probability** = add the joint probabilities of all paths that end in the effect: $P(E) = \sum_i P(C_i)P(E | C_i)$.
3. **Bayes** = one path over the total: $P(C_i | E) = \frac{P(C_i)P(E | C_i)}{P(E)}$. The posteriors over all causes add to 1 — use that as a check.
4. **Translate words carefully:** sensitivity = $P(+ | D)$; specificity = $P(- | D^c)$, so false-positive rate = $1 - \text{specificity}$; “given the alarm sounded” means the alarm is the conditioning event, not the cause.
5. **Comment questions:** compare the posterior with the prior (“the flag raised the probability from 2 % to 27 % — still less than even, so most flags are false alarms”).

Formula card

Conditional: $P(B | A) = \frac{P(A \cap B)}{P(A)}$, $P(A) > 0$

Chain: $P(A \cap B \cap C) = P(A)P(B | A)P(C | A \cap B)$

Total probability: $P(E) = \sum_i P(C_i)P(E | C_i)$, C_i a partition

Two causes: $P(E) = P(C)P(E | C) + P(C^c)P(E | C^c)$

Sensitivity = $P(+ | D)$; specificity = $P(- | D^c)$

Multiplication: $P(A \cap B) = P(A)P(B | A) = P(B)P(A | B)$

Independent $\Leftrightarrow P(B | A) = P(B)$

Bayes: $P(C_k | E) = \frac{P(C_k)P(E | C_k)}{\sum_i P(C_i)P(E | C_i)}$

Complement of a conditional: $P(B^c | A) = 1 - P(B | A)$

PPV = $P(D | +)$; NPV = $P(D^c | -)$

Question map

Q	Shape of the question	Marks
3.1	Two causes: total probability of a failure, then which cause is more likely given a failure	3
3.2	Three suppliers with defect rates: $P(\text{defective})$, all three posteriors, a "not defective and from ..." joint	5
3.3	Rare condition + diagnostic test: PPV, $P(\text{no disease} \mid \text{positive})$, $P(\text{disease} \mid \text{negative})$, interpretation	5
3.4	Conditional probabilities from a 3×2 count table; independence check	4
3.5	Chain multiplication rule: cards without replacement vs with replacement	3
3.6	Bayes with a reversed conditional ("90 % of the not-recovered had no tracker")	4
3.7	Total probability and Bayes where the second stage depends on a competitor's action	4
3.8	Recover an unknown share so that the overall rate hits a target (total probability run backwards)	3

Q 3.1 Two causes · total probability · Bayes

3 marks

A web request is routed to data centre 1 with probability 0.6 and to data centre 2 otherwise. A request fails with probability 0.02 at centre 1 and 0.05 at centre 2.

- a) Probability that a random request fails. [1.5]
 b) A request failed. Probability it was routed to centre 2. [1.5]

WHAT THIS QUESTION TESTS

The two-branch tree, total probability, and one Bayes posterior.

HOW TO ATTEMPT

- Priors 0.6 / 0.4; conditionals 0.02 / 0.05.
- $P(F) = 0.6(0.02) + 0.4(0.05)$.
- $P(D_2 \mid F)$ = its path over $P(F)$.

ANSWERS AT A GLANCE

a) 0.032 b) 0.625 (and $P(D_1 \mid F) = 0.375$)

STEP-BY-STEP SOLUTION

STEP part a Total probability

Let D_1, D_2 = routed to centre 1, 2; F = fails. $P(D_1) = 0.6$, $P(D_2) = 0.4$, $P(F \mid D_1) = 0.02$, $P(F \mid D_2) = 0.05$
 $P(F) = 0.6 \times 0.02 + 0.4 \times 0.05 = 0.012 + 0.020 = 0.032$

0.032

STEP part b Bayes

$$P(D_2 \mid F) = \frac{P(D_2)P(F \mid D_2)}{P(F)} = \frac{0.020}{0.032} = 0.625$$

Check: $P(D_1 \mid F) = 0.012/0.032 = 0.375$; $0.625 + 0.375 = 1$ ✓

0.625 — although only 40 % of traffic goes to centre 2, it accounts for 62.5 % of failures.

WHERE MARKS ARE LOST

Answering (b) with 0.05 (that is $P(F \mid D_2)$, the wrong direction) · forgetting to weight by the priors in $P(F)$.

Q 3.2 Three causes · total probability · all posteriors · joint with complement

5 marks

A phone maker buys screens from three suppliers: S1 supplies 50 %, S2 30 % and S3 20 % of the screens. Their defect rates are 4 %, 3 % and 6 % respectively. A screen is picked at random from stock.

- a) Probability that it is defective. [1.5]
- b) Given that it is defective, find the probability that it came from each supplier. Which supplier is the most likely source of a defective screen? [2.5]
- c) Probability that the screen is *not* defective and came from S3. [1]

WHAT THIS QUESTION TESTS

The three-cause tree with all posteriors, plus the difference between a posterior (conditional) and a joint probability.

HOW TO ATTEMPT

1. Compute the three joint probabilities $P(S_i \cap D)$; add for $P(D)$.
2. Each posterior = joint / $P(D)$; check they add to 1.
3. (c) is a path, not a posterior: $P(S_3)P(D^c | S_3)$.

ANSWERS AT A GLANCE

a) 0.041 b) S1 0.4878, S2 0.2195, S3 0.2927 $\Rightarrow$ S1 most likely c) 0.188

STEP-BY-STEP SOLUTION**STEP** part a Joint probabilities and total

$$P(S_1 \cap D) = 0.50 \times 0.04 = 0.020$$

$$P(S_2 \cap D) = 0.30 \times 0.03 = 0.009$$

$$P(S_3 \cap D) = 0.20 \times 0.06 = 0.012$$

$$P(D) = 0.020 + 0.009 + 0.012 = 0.041$$

0.041 (4.1 % of screens are defective)

STEP part b Posteriors by Bayes

$$P(S_1 | D) = 0.020/0.041 = 0.4878$$

$$P(S_2 | D) = 0.009/0.041 = 0.2195$$

$$P(S_3 | D) = 0.012/0.041 = 0.2927$$

$$\text{Sum} = 1.0000 \checkmark$$

S1 is the most likely source (48.8 %) — not because its rate is worst (S3's is) but because it supplies half the stock. S3 has the highest defect rate yet only 29.3 % of defectives, because it supplies only 20 %.

STEP part c A joint probability, not a posterior

$$P(S_3 \cap D^c) = P(S_3)P(D^c | S_3) = 0.20 \times 0.94 = 0.188$$

0.188

WHERE MARKS ARE LOST

Picking S3 as "most likely" by looking only at the 6 % rate · in (c) dividing by $P(D^c)$ when no conditioning is asked · rounding 0.041 to 0.04 before dividing (posteriors then do not sum to 1).

Q 3.3 Rare condition · sensitivity and specificity · PPV · interpretation

5 marks

A rare condition affects 0.5 % of the population. A screening test has sensitivity 95 % (positive when the condition is present) and specificity 90 % (negative when the condition is absent). A randomly selected person is tested.

- a) Probability that the test is positive. [1]
- b) Probability that the person has the condition given a positive result. [1.5]
- c) Probability that the person does *not* have the condition despite a positive result. [0.5]
- d) Probability that the person has the condition given a negative result. [1]
- e) Does a positive result strongly indicate the condition? Explain using your numbers. [1]

WHAT THIS QUESTION TESTS

Turning sensitivity/specificity into the four conditional probabilities, and the classic base-rate result: a good test on a rare condition still yields a low positive predictive value.

HOW TO ATTEMPT

- $P(D) = 0.005$, $P(+ | D) = 0.95$, $P(+ | D^c) = 1 - 0.90 = 0.10$.
- Total probability for $P(+)$; Bayes for (b); (c) = $1 - (b)$.
- For (d) repeat with the negative branch: $P(- | D) = 0.05$, $P(- | D^c) = 0.90$.

ANSWERS AT A GLANCE

a) 0.10425 b) 0.0456 c) 0.9544 d) 0.00028 e) No — only 4.6 % of positives are real, though the negative result is very reassuring

STEP-BY-STEP SOLUTION**STEP** part a Set up and total probability

$$P(D) = 0.005, P(D^c) = 0.995, P(+ | D) = 0.95, P(+ | D^c) = 0.10$$

$$P(+) = 0.005 \times 0.95 + 0.995 \times 0.10 = 0.00475 + 0.0995 = 0.10425$$

0.10425**STEP** part b Positive predictive value

$$P(D | +) = \frac{0.00475}{0.10425} = 0.0456$$

0.0456 (about 4.6 %)**STEP** part c False alarm given positive

$$P(D^c | +) = 1 - 0.0456 = 0.9544 \text{ (directly: } 0.0995/0.10425)$$

0.9544**STEP** part d Condition given negative

$$P(-) = 0.005 \times 0.05 + 0.995 \times 0.90 = 0.00025 + 0.8955 = 0.89575$$

$$P(D | -) = \frac{0.00025}{0.89575} = 0.00028$$

0.00028 — a negative result almost rules the condition out (NPV = 0.9997).

STEP part e Interpretation

No. The positive result raises the probability from 0.5 % to 4.6 % — a nine-fold increase, but still far below 50 %. Because the condition is rare, the 10 % false-positive rate applied to the large healthy group (0.0995) swamps the true positives (0.00475). A positive screen should be followed by a confirmatory test, not treated as a diagnosis.

WHERE MARKS ARE LOST

Using specificity 0.90 as the false-positive rate (it is 0.10) · answering (b) with 0.95 ("the test is 95 % accurate") · leaving (e) as a vague sentence without the 0.5 % → 4.6 % comparison.

Q 3.4 Count table · conditional probability · independence

4 marks

A lender classified 1000 past loans by borrower age and default status:

	Defaulted	Repaid	Total
Young	60	240	300
Middle-aged	45	405	450
Senior	15	235	250
Total	120	880	1000

- a) Probability that a borrower defaulted given they are young. [1]
- b) Probability that a borrower is young given they defaulted. [1]
- c) Probability that a borrower is young or defaulted. [1]
- d) Are "Young" and "Defaulted" independent? Which age group has the lowest default rate? [1]

WHAT THIS QUESTION TESTS

Row-conditional vs column-conditional, the addition rule from a table, and an independence check.

HOW TO ATTEMPT

- Given Young $\Rightarrow$ divide by 300. Given Defaulted $\Rightarrow$ divide by 120.
- Union = $P(Y) + P(D) - P(Y \cap D)$.
- Compare $P(D | Y)$ with $P(D)$; compute the three row rates.

ANSWERS AT A GLANCE

a) 0.20 b) 0.50 c) 0.36 d) Not independent ($0.20 \neq 0.12$); seniors lowest at 0.06

STEP-BY-STEP SOLUTION**STEP** part a Given young

$$P(D | Y) = 60/300 = 0.20$$

STEP part b Given defaulted

$$P(Y | D) = 60/120 = 0.50$$

STEP part c Union

$$P(Y) = 0.30, P(D) = 0.12, P(Y \cap D) = 60/1000 = 0.06$$

$$P(Y \cup D) = 0.30 + 0.12 - 0.06 = 0.36$$

0.36

STEP **part d** Independence and rates

$P(D | Y) = 0.20 \neq P(D) = 0.12 \Rightarrow$ dependent (also $0.30 \times 0.12 = 0.036 \neq 0.06$)

Rates: Young $60/300 = 0.20$, Middle $45/450 = 0.10$, Senior $15/250 = 0.06$

Dependent — young borrowers default at 20 %, well above the overall 12 %. Seniors have the lowest rate, 6 %.

WHERE MARKS ARE LOST

Mixing up the denominators in (a) and (b) · using the count 60 as if it were a probability · claiming independence because both conditionals “exist”.

Q 3.5 Chain multiplication rule · with and without replacement**3 marks**

From a well-shuffled standard deck of 52 cards, cards are drawn one at a time.

- a) Three cards are drawn without replacement. Probability that all three are aces. [1]
- b) Two cards are drawn without replacement. Probability that the first is a king and the second a queen. [1]
- c) Repeat (a) with replacement. [1]

WHAT THIS QUESTION TESTS

The chain rule $P(A \cap B \cap C) = P(A)P(B | A)P(C | A \cap B)$ and how the denominator shrinks without replacement.

HOW TO ATTEMPT

- Without replacement: numerators and denominators both drop by one each draw.
- With replacement: independent draws, same probability each time.

ANSWERS AT A GLANCE

a) $1/5525 = 0.000181$ b) $4/663 = 0.00603$ c) $1/2197 = 0.000455$

STEP-BY-STEP SOLUTION**STEP** **part a** Three aces without replacement

$$P = \frac{4}{52} \times \frac{3}{51} \times \frac{2}{50} = \frac{24}{132600} = \frac{1}{5525} = 0.000181$$

STEP **part b** King then queen

$$P = \frac{4}{52} \times \frac{4}{51} = \frac{16}{2652} = \frac{4}{663} = 0.00603$$

STEP **part c** With replacement

$$P = \left(\frac{4}{52}\right)^3 = \left(\frac{1}{13}\right)^3 = \frac{1}{2197} = 0.000455$$

With replacement the probability is larger (0.000455 vs 0.000181) because each draw still has all four aces available.

WHERE MARKS ARE LOST

Keeping the denominator 52 in every factor without replacement · in (b) reducing the number of queens after a king is drawn (queens are untouched: still 4 of 51).

Q 3.6 Bayes · reversed conditional in the wording

4 marks

Of the parcels reported lost by a courier, 70 % are eventually recovered. Among recovered parcels, 60 % had a GPS tracker. Among parcels never recovered, 90 % had *no* tracker. A parcel has just been reported lost.

- a) If it has a tracker, find the probability that it will *not* be recovered. [2]
 b) If it has no tracker, find the probability that it will be recovered. [2]

WHAT THIS QUESTION TESTS

Reading "90 % of the not-recovered had no tracker" as $P(T^c | R^c) = 0.9$, hence $P(T | R^c) = 0.1$, then Bayes in both directions.

HOW TO ATTEMPT

- Causes: R (recovered, 0.7) and R^c (0.3). Effect: T (tracker). $P(T | R) = 0.6$, $P(T | R^c) = 0.1$.
- Four path probabilities; then the two requested posteriors.

ANSWERS AT A GLANCE

a) $0.03/0.45 = 0.0667$ b) $0.28/0.55 = 0.5091$

STEP-BY-STEP SOLUTION**STEP** Path probabilities

$$P(R \cap T) = 0.7 \times 0.6 = 0.42; P(R \cap T^c) = 0.7 \times 0.4 = 0.28$$

$$P(R^c \cap T) = 0.3 \times 0.1 = 0.03; P(R^c \cap T^c) = 0.3 \times 0.9 = 0.27$$

$$\text{Check: } 0.42 + 0.28 + 0.03 + 0.27 = 1 \checkmark; P(T) = 0.45, P(T^c) = 0.55$$

STEP part a Not recovered given tracker

$$P(R^c | T) = \frac{0.03}{0.45} = 0.0667$$

0.0667 — a tracked parcel is recovered 93.3 % of the time.

STEP part b Recovered given no tracker

$$P(R | T^c) = \frac{0.28}{0.55} = 0.5091$$

0.5091 — without a tracker recovery is close to a coin toss.

WHERE MARKS ARE LOST

Using 0.9 as $P(T | R^c)$ instead of $P(T^c | R^c)$ · dividing by 0.3 (the prior) instead of by $P(T)$ · reporting $P(T | R^c)$ when $P(R^c | T)$ is asked.

Q 3.7 Total probability · Bayes · second stage depends on a competitor

4 marks

A start-up applies for a government grant. The probability that a rival firm also applies is 0.6. If the rival applies, the start-up wins with probability 0.3; if the rival does not apply, the start-up wins with probability 0.8.

- a) Probability that the start-up wins the grant. [1.5]
 b) Given that the start-up won, probability that the rival did not apply. [1.5]
 c) Given that the start-up lost, probability that the rival applied. [1]

WHAT THIS QUESTION TESTS

A total-probability question where the “cause” is another party’s action, and Bayes for both outcomes.

HOW TO ATTEMPT

1. $P(A) = 0.6$ rival applies; $P(W | A) = 0.3$, $P(W | A^c) = 0.8$.
2. Paths, total, then posteriors.

ANSWERS AT A GLANCE

a) 0.50 b) 0.64 c) 0.84

STEP-BY-STEP SOLUTION**STEP** part a **Total probability**

$$P(W) = 0.6 \times 0.3 + 0.4 \times 0.8 = 0.18 + 0.32 = 0.50$$

0.50

STEP part b **Rival absent given win**

$$P(A^c | W) = \frac{0.32}{0.50} = 0.64$$

0.64

STEP part c **Rival present given loss**

$$P(L) = 0.50; P(A \cap L) = 0.6 \times 0.7 = 0.42$$

$$P(A | L) = \frac{0.42}{0.50} = 0.84$$

0.84

WHERE MARKS ARE LOST

Using 0.3 and 0.8 as if they were unconditional · in (c) forgetting that $P(L | A) = 1 - 0.3 = 0.7$.

Q 3.8 Total probability run backwards · recover a share

3 marks

Three plants make the same component. Plant A makes 40 % of output and 98 % of its components pass inspection. Plants B and C share the remaining 60 %; their pass rates are 94 % and 97 % respectively. What share of total output should B and C each produce so that a randomly chosen component passes with probability exactly 96 %?

WHAT THIS QUESTION TESTS

Setting up total probability with an unknown share and solving a linear equation — a “recover the parameter” version of the usual question.

HOW TO ATTEMPT

1. Let B’s share be b ; then C’s share is $0.6 - b$.
2. Write $P(\text{pass}) = 0.4(0.98) + b(0.94) + (0.6 - b)(0.97) = 0.96$; solve.
3. Sanity-check: overall 96 % lies between B’s 94 % and C’s 97 %, so a mix is possible.

ANSWERS AT A GLANCE

B: 46.7 % of output, C: 13.3 %

STEP-BY-STEP SOLUTION

STEP Equation

$$0.4(0.98) + 0.94b + 0.97(0.6 - b) = 0.96$$

$$0.392 + 0.94b + 0.582 - 0.97b = 0.96$$

$$0.974 - 0.03b = 0.96 \Rightarrow 0.03b = 0.014 \Rightarrow b = 0.4667$$

$$\text{C's share} = 0.6 - 0.4667 = 0.1333$$

$$\text{Check: } 0.392 + 0.4667(0.94) + 0.1333(0.97) = 0.392 + 0.4387 + 0.1293 = 0.960 \checkmark$$

B should make 46.7 % and C 13.3 % of total output.

WHERE MARKS ARE LOST

Treating B and C as 30 % each and "checking" instead of solving · algebra slips with the $0.97(0.6 - b)$ term · giving b without C's share.

UNIT 4

Naïve Bayes Classification

Frequency tables from raw data · priors and likelihoods · the conditional-independence assumption · comparing posteriors · zero-frequency problem and Laplace smoothing · text (bag-of-words) classification

What this unit is about

Naïve Bayes is **Bayes' theorem used as a classifier**. For a new record with features $x_1, \dots, x_n$ and a candidate class C , we want $P(C \mid x_1, \dots, x_n)$. Bayes says this is proportional to $P(C) P(x_1, \dots, x_n \mid C)$. The "naïve" step assumes the features are independent given the class, so the likelihood becomes a simple product: $P(C) \prod_i P(x_i \mid C)$. Compute this score for every class and pick the largest — that is the whole algorithm.

Two details carry marks. First, the denominator $P(x_1, \dots, x_n)$ is the same for every class, so you may compare unnormalised scores; but if the question asks for *posterior probabilities*, divide each score by the sum of the scores. Second, if a feature value never appears with some class, its likelihood is 0 and the whole product collapses — the **zero-frequency problem**. **Laplace smoothing** fixes it by adding 1 to every count (and the number of categories, or the vocabulary size, to the denominator).

How to attempt questions from this unit

- Step 1 – frequency table.** Convert the raw rows into counts of each feature value within each class. Write the class totals.
- Step 2 – priors.** $P(C) = \text{class count} \div \text{total rows}$.
- Step 3 – likelihoods** for the *new* record only: $P(x_i \mid C) = \text{count of } x_i \text{ in class} \div \text{class count}$. You do not need the whole table of likelihoods — only the values that appear in the new record.
- Step 4 – scores.** $\text{Score}(C) = P(C) \prod_i P(x_i \mid C)$. Keep fractions or 4 decimals.
- Step 5 – decide**, and if asked, normalise: $\text{posterior} = \text{score} \div (\text{sum of scores})$.
- Text data:** vocabulary V = number of distinct words in the training set; $P(w \mid C) = \frac{\text{count}(w, C) + 1}{\text{total words in } C + |V|}$ with Laplace smoothing; a repeated word in the message is multiplied as many times as it appears.

Formula card

$$\text{Bayes: } P(C \mid X) = \frac{P(X \mid C)P(C)}{P(X)}$$

$$\text{Decision: } \hat{C} = \arg \max_C P(C) \prod_i P(x_i \mid C)$$

$$\text{Laplace (categorical, } k \text{ values): } \frac{n_{x,C} + 1}{n_C + k}$$

Posteriors over all classes sum to 1

$$\text{Naïve assumption: } P(x_1, \dots, x_n \mid C) = \prod_i P(x_i \mid C)$$

$$\text{Posterior: } \text{score}(C) \div \sum \text{score}(\text{all classes})$$

$$\text{Laplace (text): } \frac{\text{count}(w, C) + 1}{N_C + |V|}$$

Same denominator $\Rightarrow$ compare numerators only

Question map

Q	Shape of the question	Marks
4.1	Single feature from raw rows: frequency table, priors, likelihoods, posterior for one value	5
4.2	Two features from a summary count table: priors, likelihoods, normalised posteriors, decision	5
4.3	Three features from raw rows (10 records): full Naïve Bayes prediction	5
4.4	Text classification with Laplace smoothing, unseen word	5
4.5	Zero-frequency problem from a word-count table, before and after smoothing	5

Q 4.1 Single feature · frequency table · posterior

5 marks

A gym recorded the weather and whether a member turned up for the morning session:

#	Weather	Attended	#	Weather	Attended
1	Sunny	Yes	7	Rainy	No
2	Rainy	No	8	Sunny	Yes
3	Cloudy	Yes	9	Cloudy	No
4	Sunny	No	10	Rainy	Yes
5	Cloudy	Yes	11	Sunny	Yes
6	Sunny	Yes	12	Rainy	No

- a) Build the frequency table (Weather $\times$ Attended) and state the prior probabilities of Yes and No. [2]
- b) Tomorrow is Cloudy. Using Bayes' theorem, compute $P(\text{Yes} \mid \text{Cloudy})$ and $P(\text{No} \mid \text{Cloudy})$ and predict whether the member attends. [3]

WHAT THIS QUESTION TESTS

Counting correctly from raw data, and the full Bayes computation with normalisation for a single feature.

HOW TO ATTEMPT

- Tally the 12 rows into a 3×2 table with row and column totals.
- Priors from the column totals.
- Likelihoods $P(\text{Cloudy} \mid \text{Yes})$ and $P(\text{Cloudy} \mid \text{No})$; scores; normalise by $P(\text{Cloudy})$.

ANSWERS AT A GLANCE

- a) Yes 7, No 5; $P(\text{Yes}) = 7/12$, $P(\text{No}) = 5/12$
- b) $P(\text{Yes} \mid \text{Cloudy}) = 2/3$, $P(\text{No} \mid \text{Cloudy}) = 1/3 \Rightarrow$ predict Yes

STEP-BY-STEP SOLUTION

STEP part a Frequency table and priors

Weather	Yes	No	Total
Sunny	4	1	5
Cloudy	2	1	3
Rainy	1	3	4
Total	7	5	12

$$P(\text{Yes}) = 7/12 = 0.5833, \quad P(\text{No}) = 5/12 = 0.4167$$

STEP **part b** Likelihoods and scores

$$P(\text{Cloudy} \mid \text{Yes}) = 2/7; P(\text{Cloudy} \mid \text{No}) = 1/5$$

$$\text{Score}(\text{Yes}) = P(\text{Yes})P(\text{Cloudy} \mid \text{Yes}) = \frac{7}{12} \times \frac{2}{7} = \frac{2}{12}$$

$$\text{Score}(\text{No}) = \frac{5}{12} \times \frac{1}{5} = \frac{1}{12}$$

$$P(\text{Cloudy}) = \frac{2}{12} + \frac{1}{12} = \frac{3}{12} \text{ (matches the row total } 3/12 \text{ ✓)}$$

STEP **part b** Posteriors and decision

$$P(\text{Yes} \mid \text{Cloudy}) = \frac{2/12}{3/12} = \frac{2}{3} = 0.6667$$

$$P(\text{No} \mid \text{Cloudy}) = \frac{1/12}{3/12} = \frac{1}{3} = 0.3333$$

Since $0.6667 > 0.3333$, predict that the member attends.

WHERE MARKS ARE LOST

Miscounting rows (write the tally marks) · using $P(\text{Yes} \mid \text{Cloudy}) = 2/3$ straight from the row and skipping the Bayes working (the exam asks you to show it) · forgetting the prior and comparing $2/7$ with $1/5$ only.

Q 4.2 Two features · summary counts · normalised posteriors

5 marks

An ad platform summarised past impressions by device, time of day and whether the user clicked:

Device	Time	Click = Yes	Click = No
Mobile	Day	20	30
Mobile	Night	30	10
Desktop	Day	10	40
Desktop	Night	10	50

A new impression is served on **Mobile** at **Night**. Using Naïve Bayes:

- Compute the prior probabilities of the two classes. [1]
- Compute the likelihoods $P(\text{Mobile} \mid \text{class})$ and $P(\text{Night} \mid \text{class})$ for both classes. [1]
- Compute the posterior probabilities of Click = Yes and Click = No. [2]
- Predict the outcome and interpret for the marketing team. [1]

WHAT THIS QUESTION TESTS

Extracting per-feature counts from a two-feature summary table (add rows), applying the independence assumption, and normalising.

HOW TO ATTEMPT

- Class totals: Yes = $20+30+10+10 = 70$, No = $30+10+40+50 = 130$; total 200.
- Mobile within Yes = $20+30 = 50$; Night within Yes = $30+10 = 40$. Same for No.
- Score = prior $\times$ two likelihoods; normalise.

ANSWERS AT A GLANCE

- a) $P(Y) = 0.35$, $P(N) = 0.65$ b) $P(M \mid Y) = 5/7$, $P(Ni \mid Y) = 4/7$, $P(M \mid N) = 4/13$, $P(Ni \mid N) = 6/13$ c) Yes 0.6075, No 0.3925 d) Click

STEP-BY-STEP SOLUTION

STEP **part a** **Priors**

Yes total = $20 + 30 + 10 + 10 = 70$; No total = $30 + 10 + 40 + 50 = 130$; overall 200

$P(\text{Yes}) = 70/200 = 0.35$, $P(\text{No}) = 130/200 = 0.65$

STEP **part b** **Likelihoods for Mobile and Night**

Within Yes: Mobile = $20 + 30 = 50$, Night = $30 + 10 = 40$

$P(\text{Mobile} | \text{Yes}) = 50/70 = 5/7 = 0.7143$; $P(\text{Night} | \text{Yes}) = 40/70 = 4/7 = 0.5714$

Within No: Mobile = $30 + 10 = 40$, Night = $10 + 50 = 60$

$P(\text{Mobile} | \text{No}) = 40/130 = 4/13 = 0.3077$; $P(\text{Night} | \text{No}) = 60/130 = 6/13 = 0.4615$

STEP **part c** **Scores and posteriors**

Score(Yes) = $0.35 \times \frac{5}{7} \times \frac{4}{7} = 0.35 \times 0.40816 = 0.14286$

Score(No) = $0.65 \times \frac{4}{13} \times \frac{6}{13} = 0.65 \times 0.14201 = 0.09231$

Sum = 0.23516

$P(\text{Yes} | \text{Mobile, Night}) = 0.14286/0.23516 = 0.6075$; $P(\text{No} | \cdot) = 0.09231/0.23516 = 0.3925$

Posteriors: Yes 0.6075, No 0.3925

STEP **part d** **Decision**

Predict Click = Yes (about 61 %). Even though only 35 % of impressions are clicked overall, the Mobile-at-Night combination is strongly associated with clicking; the team should prioritise night-time mobile placements.

WHERE MARKS ARE LOST

Using the single cell (Mobile, Night) = 30/70 as the likelihood — that is not the Naïve Bayes method the question asks for · forgetting to add rows to get per-feature counts · reporting unnormalised scores as “posterior probabilities”.

Q 4.3 Three features · raw rows · full prediction

5 marks

Ten past days of a cricket ground's decision to hold practice:

Day	Outlook	Temp	Wind	Practice
1	Sunny	Hot	Weak	No
2	Sunny	Hot	Strong	No
3	Overcast	Hot	Weak	Yes
4	Rain	Mild	Weak	Yes
5	Rain	Cool	Weak	Yes
6	Rain	Cool	Strong	No
7	Overcast	Cool	Strong	Yes
8	Sunny	Mild	Weak	No
9	Sunny	Cool	Weak	Yes
10	Rain	Mild	Strong	Yes

Using the Naïve Bayes classifier, predict whether practice is held on a day that is **Sunny, Cool, Strong**. Show priors, likelihoods, scores and the posterior probabilities.

WHAT THIS QUESTION TESTS

The complete multi-feature workflow on raw data, including the discipline of only computing the likelihoods needed.

HOW TO ATTEMPT

1. Class counts: Yes = days 3,4,5,7,9,10 = 6; No = days 1,2,6,8 = 4.
2. Within Yes count Sunny, Cool, Strong; within No the same.
3. Multiply; normalise; decide.

ANSWERS AT A GLANCE

Score(Yes) = 0.01667, Score(No) = 0.0375 $\Rightarrow$ posteriors Yes 0.308, No 0.692 $\Rightarrow$ **No practice**

STEP-BY-STEP SOLUTION**STEP Priors**

Yes: days 3, 4, 5, 7, 9, 10 $\rightarrow$ 6; No: days 1, 2, 6, 8 $\rightarrow$ 4

$$P(\text{Yes}) = 6/10 = 0.6, \quad P(\text{No}) = 4/10 = 0.4$$

STEP Likelihoods needed for (Sunny, Cool, Strong)

Feature value	Count in Yes (of 6)	$P(\cdot \text{Yes})$	Count in No (of 4)	$P(\cdot \text{No})$
Sunny	1 (day 9)	1/6	3 (days 1, 2, 8)	3/4
Cool	3 (days 5, 7, 9)	3/6	1 (day 6)	1/4
Strong	2 (days 7, 10)	2/6	2 (days 2, 6)	2/4

STEP Scores

$$\text{Score(Yes)} = 0.6 \times \frac{1}{6} \times \frac{3}{6} \times \frac{2}{6} = 0.6 \times \frac{6}{216} = 0.6 \times 0.02778 = 0.01667$$

$$\text{Score(No)} = 0.4 \times \frac{3}{4} \times \frac{1}{4} \times \frac{2}{4} = 0.4 \times \frac{6}{64} = 0.4 \times 0.09375 = 0.03750$$

STEP Posteriors and decision

$$\text{Sum} = 0.05417$$

$$P(\text{Yes} | X) = 0.01667/0.05417 = 0.308; P(\text{No} | X) = 0.03750/0.05417 = 0.692$$

Score(No) > Score(Yes) $\Rightarrow$ predict No practice (posterior 0.692). The Sunny days in the data were mostly No-days, and that outweighs the higher prior for Yes.

WHERE MARKS ARE LOST

Counting a feature value across all 10 rows instead of within each class · dividing Sunny-within-Yes by 10 instead of by 6 · arithmetic in the products (write the fraction, then the decimal).

Q 4.4 Text classification · bag of words · Laplace smoothing

5 marks

A tiny SMS data set:

Message	Label
win cash prize now	Spam
cash offer limited	Spam
meeting at noon	Ham
send the report now	Ham

- a) List the vocabulary and give its size $|V|$, the priors, and the total word count in each class. [1]
- b) Using Naïve Bayes with Laplace smoothing ($\alpha = 1$), classify the new message “cash prize meeting”. Give the posterior probability of each class. [3]
- c) What would happen without smoothing? Answer with the two scores. [1]

WHAT THIS QUESTION TESTS

The bag-of-words likelihood with smoothing, and why smoothing is essential when a word is absent from a class.

HOW TO ATTEMPT

- Vocabulary = distinct words across all four messages.
- $P(w | C) = \frac{\text{count} + 1}{N_C + |V|}$ for each of the three words in the new message, per class.
- Score = prior $\times$ product of three likelihoods; normalise.

ANSWERS AT A GLANCE

- a) $|V| = 12$; priors $0.5 / 0.5$; 7 words in Spam, 7 in Ham
- b) $\text{Score}(\text{Spam}) = 0.5 \times \frac{3}{19} \cdot \frac{2}{19} \cdot \frac{1}{19}$, $\text{Score}(\text{Ham}) = 0.5 \times \frac{1}{19} \cdot \frac{1}{19} \cdot \frac{2}{19} \Rightarrow$ posteriors Spam 0.75, Ham 0.25 $\Rightarrow$ Spam
- c) Both scores become 0 — no decision possible

STEP-BY-STEP SOLUTION**STEP** part a Vocabulary, priors, word totals

Vocabulary: win, cash, prize, now, offer, limited, meeting, at, noon, send, the, report $\rightarrow |V| = 12$

Priors: 2 Spam of 4, 2 Ham of 4 $\Rightarrow P(S) = P(H) = 0.5$

Total words: Spam = 4 + 3 = 7; Ham = 3 + 4 = 7

STEP part b Smoothed likelihoods (denominator 7 + 12 = 19)

Word	Count in Spam	$P(w S)$	Count in Ham	$P(w H)$
cash	2	$(2+1)/19 = 3/19$	0	$(0+1)/19 = 1/19$
prize	1	$2/19$	0	$1/19$
meeting	0	$1/19$	1	$2/19$

STEP **part b** Scores and posteriors

$$\text{Score}(\text{Spam}) = 0.5 \times \frac{3 \times 2 \times 1}{19^3} = 0.5 \times \frac{6}{6859} = 4.374 \times 10^{-4}$$

$$\text{Score}(\text{Ham}) = 0.5 \times \frac{1 \times 1 \times 2}{19^3} = 0.5 \times \frac{2}{6859} = 1.458 \times 10^{-4}$$

$$\text{Ratio } 6 : 2 = 3 : 1$$

$$P(\text{Spam} \mid \text{msg}) = \frac{6}{6+2} = 0.75, P(\text{Ham} \mid \text{msg}) = 0.25$$

Classify as Spam (posterior 0.75).

STEP **part c** Without smoothing

$$\text{Score}(\text{Spam}) = 0.5 \times \frac{2}{7} \times \frac{1}{7} \times \frac{0}{7} = 0; \text{Score}(\text{Ham}) = 0.5 \times \frac{0}{7} \times \frac{0}{7} \times \frac{1}{7} = 0$$

Both scores are 0 because each class is missing at least one word of the message — the classifier cannot decide. Smoothing prevents this collapse.

WHERE MARKS ARE LOST

Using $|V| = 4$ (number of messages) instead of 12 · forgetting the +1 in the numerator but adding $|V|$ in the denominator · counting “now” (appears in both classes) — irrelevant, it is not in the new message.

Q 4.5 Zero-frequency problem · word-count table · smoothing changes the decision**5 marks**

Word counts in a company's past e-mails, by class, are summarised below. From past volume, 60 % of e-mails are Normal and 40 % are Spam.

Word	Normal	Spam
hello	6	2
meeting	4	0
invoice	2	3
free	0	5
Total	12	10

- a) Without smoothing, compute the Naïve Bayes scores for the message “free meeting”. What goes wrong? [1.5]
- b) Apply Laplace smoothing (add 1 to every count) and rebuild the table. [1.5]
- c) Recompute the scores and posteriors, and classify the message. [2]

WHAT THIS QUESTION TESTS

Recognising the zero-frequency collapse and applying smoothing to a categorical count table (add 1 per cell, add the number of words to each class total).

HOW TO ATTEMPT

1. Likelihood of each word = its count ÷ class total.
2. Smoothing: each cell +1; totals become $12+4 = 16$ and $10+4 = 14$.
3. Scores with the given priors 0.6 / 0.4; normalise.

ANSWERS AT A GLANCE

- a) Both scores 0 (free never in Normal, meeting never in Spam)
- b) Normal totals 16, Spam totals 14
- c) $\text{Score}(\text{Normal}) = 0.01172$, $\text{Score}(\text{Spam}) = 0.01224 \Rightarrow$ posteriors 0.489 / 0.511 $\Rightarrow$ Spam (narrowly)

STEP-BY-STEP SOLUTION

STEP part a Unsmoothed

$$\text{Score(Normal)} = 0.6 \times P(\text{free} \mid N) P(\text{meeting} \mid N) = 0.6 \times \frac{0}{12} \times \frac{4}{12} = 0$$

$$\text{Score(Spam)} = 0.4 \times \frac{5}{10} \times \frac{0}{10} = 0$$

Both scores are zero. One unseen word wipes out all the evidence from the other words, and no classification is possible.

STEP part b Smoothed table

Word	Normal (+1)	Spam (+1)
hello	7	3
meeting	5	1
invoice	3	4
free	1	6
Total	16	14

STEP part c Smoothed scores

$$\text{Score(Normal)} = 0.6 \times \frac{1}{16} \times \frac{5}{16} = 0.6 \times 0.019531 = 0.011719$$

$$\text{Score(Spam)} = 0.4 \times \frac{6}{14} \times \frac{1}{14} = 0.4 \times 0.030612 = 0.012245$$

$$\text{Sum} = 0.023964; \text{posteriors: Normal } 0.011719/0.023964 = 0.489, \text{ Spam } 0.511$$

Classify as Spam, but only just (51.1 % vs 48.9 %) — “free” pulls towards Spam, “meeting” pulls towards Normal, and the Normal prior nearly balances them. Report the closeness in your interpretation.

WHERE MARKS ARE LOST

Adding 1 only to the zero cells (every cell gets +1) · leaving the totals at 12 and 10 after smoothing · using word-count totals (12/22, 10/22) as priors when the question gives 60 % / 40 %.

UNIT 5

Discrete Random Variables

pmf and its two conditions · finding an unknown constant · expectation, variance, $E(X^2)$ · linear transformations · CDF · joint pmf tables and formulas · marginals · conditional probabilities · independence · covariance

What this unit is about

A random variable is a **number attached to each outcome**. When the possible values can be listed (0, 1, 2, ...) it is discrete, and its probability mass function (pmf) is simply a table: value $\rightarrow$ probability. The two rules are $p(x) \geq 0$ and $\sum p(x) = 1$; the second rule is how you find an unknown constant k . Once the table is known, everything else is arithmetic on it: the mean is the probability-weighted average of the values, the variance is $E(X^2) - [E(X)]^2$, and any probability is the sum of the relevant entries.

Joint distributions are the same idea with two variables: a two-way table whose entries add to 1. Row sums give the marginal pmf of X , column sums the marginal pmf of Y . A conditional probability such as $P(Y > 10 \mid X = 2)$ is “the part of row $X = 2$ that has $Y > 10$, divided by the whole row”. Independence means every cell equals row total $\times$ column total — a single failing cell breaks it.

How to attempt questions from this unit

- Unknown constant:** write out every $p(x)$ in terms of k , add, set equal to 1, solve (reject a negative root).
- Expectation table:** columns x , $p(x)$, $x p(x)$, $x^2 p(x)$; sum the last two to get $E(X)$ and $E(X^2)$; then $\text{Var}(X) = E(X^2) - [E(X)]^2$.
- Linear transformations:** $E(aX + b) = aE(X) + b$, $\text{Var}(aX + b) = a^2 \text{Var}(X)$, $\text{SD}(aX + b) = |a| \text{SD}(X)$. The constant b never affects the variance.
- Probabilities of ranges:** list the values that satisfy the inequality (strict vs non-strict!) and add their probabilities. $F(x) = P(X \leq x)$ accumulates from the left.
- Joint tables:** first write the row and column totals in the margins. Then: joint probability = one cell; marginal = a margin; conditional = cells $\div$ the margin you are conditioning on; independence = check cell = row $\times$ column for every cell (one counter-example is enough to say “not independent”).
- Joint pmf given by a formula:** build the table of $f(x, y)$ values first (with c factored out), then proceed as with any table.

Formula cardValid pmf: $p(x) \geq 0$, $\sum_x p(x) = 1$

$$E(X) = \sum x p(x), E(X^2) = \sum x^2 p(x)$$

$$E(aX + b) = aE(X) + b$$

Joint: $f(x, y) = P(X = x, Y = y)$, all cells sum to 1

$$\text{Conditional: } P(Y = y | X = x) = \frac{f(x, y)}{f_X(x)}$$

$$E(XY) = \sum_x \sum_y xy f(x, y)$$

$$\text{Without replacement (hypergeometric): } P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

$$\text{CDF: } F(x) = P(X \leq x) = \sum_{t \leq x} p(t)$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2, \text{SD} = \sqrt{\text{Var}}$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

$$\text{Marginals: } f_X(x) = \sum_y f(x, y), f_Y(y) = \sum_x f(x, y)$$

$$\text{Independent} \Leftrightarrow f(x, y) = f_X(x)f_Y(y) \text{ for all cells}$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) (= 0 \text{ if independent})$$

$$\text{Fair game} \Leftrightarrow \text{expected gain} = 0$$

Question map

Q	Shape of the question	Marks
5.1	pmf $k(2x + 1)$: find k , mean, variance, range probability, CDF value	5
5.2	pmf table with k and k^2 : quadratic for k , several probabilities, mean	4
5.3	$\text{Var}(aX + b)$ and probability within one SD of the mean	4
5.4	Game with gains and losses: expected gain, fairness, entry fee, variance	4
5.5	Number of defectives when sampling without replacement: full distribution, mean, CDF	4
5.6	Joint pmf table: marginals, joint and conditional probabilities, independence, $E(XY)$, covariance	6
5.7	Joint pmf by formula $c(x + y)$: constant, marginals, conditional, independence	5
5.8	Joint distribution built from an experiment (three coin tosses): table, marginals, $E(XY)$	5

Q 5.1 pmf with unknown k · mean · variance · probabilities

5 marks

A discrete random variable X has pmf $p(x) = k(2x + 1)$ for $x = 0, 1, 2, 3$ and 0 otherwise.

- a) Find k . [1]
- b) Find $E(X)$ and $E(X^2)$. [1.5]
- c) Find $\text{Var}(X)$ and the standard deviation. [1]
- d) Find $P(1 \leq X < 3)$ and $F(2) = P(X \leq 2)$. [1.5]

WHAT THIS QUESTION TESTS

The standard pmf workflow; careful reading of strict and non-strict inequalities.

HOW TO ATTEMPT

- Write the four probabilities: $k, 3k, 5k, 7k$; sum = $16k$.
- Expectation table.
- $1 \leq X < 3$ means $X \in \{1, 2\}$; $X \leq 2$ means $\{0, 1, 2\}$.

ANSWERS AT A GLANCEa) $k = 1/16$ b) $E(X) = 2.125$, $E(X^2) = 5.375$ c) $\text{Var} 0.8594$, $\text{SD} 0.927$ d) 0.5 and 0.5625 **STEP-BY-STEP SOLUTION**

STEP **part a** Total probability = 1

$$p(0) = k, p(1) = 3k, p(2) = 5k, p(3) = 7k$$

$$k + 3k + 5k + 7k = 16k = 1 \Rightarrow k = 1/16$$

$$k = 1/16$$

STEP **part b** Expectation table

x	$p(x)$	$x p(x)$	$x^2 p(x)$
0	1/16	0	0
1	3/16	3/16	3/16
2	5/16	10/16	20/16
3	7/16	21/16	63/16
Sum	1	34/16	86/16

$$E(X) = 34/16 = 2.125; E(X^2) = 86/16 = 5.375$$

STEP **part c** Variance

$$\text{Var}(X) = 5.375 - 2.125^2 = 5.375 - 4.5156 = 0.8594$$

$$\text{SD} = \sqrt{0.8594} = 0.927$$

$$\text{Var } 0.8594, \text{SD } 0.927$$

STEP **part d** Range probabilities

$$P(1 \leq X < 3) = p(1) + p(2) = 3/16 + 5/16 = 8/16 = 0.5$$

$$F(2) = p(0) + p(1) + p(2) = 9/16 = 0.5625$$

$$0.5 \text{ and } 0.5625$$

WHERE MARKS ARE LOST

Including $x = 3$ in $X < 3$ · computing variance as $\sum (x - \mu)^2$ without weighting by $p(x)$ · forgetting to square-root for the SD.

Q 5.2 pmf table with k and k^2 · quadratic in k

4 marks

A random variable X has the probability function

x	0	1	2	3	4	5
$p(x)$	k	k	k	k	$2k^2$	$3k^2$

- a) Find k . [1.5]
- b) Evaluate $P(X < 4)$ and $P(X \geq 4)$. [1]
- c) Evaluate $P(1 < X \leq 4)$. [0.5]
- d) Find $E(X)$. [1]

WHAT THIS QUESTION TESTS

Solving a quadratic for k and rejecting the invalid root; adding the right cells for each inequality.

HOW TO ATTEMPT

- Sum: $4k + 5k^2 = 1$; quadratic formula; keep the positive root.
- Fill numeric probabilities, then add.

ANSWERS AT A GLANCEa) $k = 0.2$ (reject -1) b) 0.8 and 0.2 c) 0.48 d) 2.12**STEP-BY-STEP SOLUTION****STEP** part a Quadratic

$$4k + 2k^2 + 3k^2 = 1 \Rightarrow 5k^2 + 4k - 1 = 0$$

$$k = \frac{-4 \pm \sqrt{16 + 20}}{10} = \frac{-4 \pm 6}{10} \Rightarrow k = 0.2 \text{ or } k = -1$$

 $k = -1$ gives negative probabilities $\Rightarrow$ rejected

 $k = 0.2$; pmf: 0.2, 0.2, 0.2, 0.2, 0.08, 0.12 (sum 1 ✓)
STEP part b Complementary probabilities

$$P(X < 4) = p(0) + p(1) + p(2) + p(3) = 0.8; P(X \geq 4) = 1 - 0.8 = 0.2$$

STEP part c Strict on the left, inclusive on the right

$$P(1 < X \leq 4) = p(2) + p(3) + p(4) = 0.2 + 0.2 + 0.08 = 0.48$$

STEP part d Mean

$$E(X) = 0(0.2) + 1(0.2) + 2(0.2) + 3(0.2) + 4(0.08) + 5(0.12) = 0.2 + 0.4 + 0.6 + 0.32 + 0.6 = 2.12$$

2.12

WHERE MARKS ARE LOSTNot rejecting the negative root explicitly · $1 < X \leq 4$ read as including 1 · $P(X \geq 4)$ computed as $p(4)$ only.**Q 5.3** Variance of a linear transformation · within one SD

4 marks

The number of support tickets X raised per hour has the distribution

x	2	5	8	12
$P(X = x)$	0.2	0.3	0.4	0.1

Each ticket costs ₹2 (hundred) to resolve and the desk has a fixed hourly saving of ₹3 (hundred), so the net hourly cost is $Y = 2X - 3$.

- a) Find $E(X)$, $\text{Var}(X)$ and $\text{SD}(X)$. [1.5]
- b) Find $E(Y)$, $\text{Var}(Y)$ and $\text{SD}(Y)$. [1.5]
- c) Find the probability that X lies within one standard deviation of its mean, and interpret. [1]

WHAT THIS QUESTION TESTSThe rules $E(aX + b) = aE(X) + b$ and $\text{Var}(aX + b) = a^2\text{Var}(X)$, and reading a probability off a discrete table for an interval.**HOW TO ATTEMPT**

- Expectation table for X .
- Apply the linear rules (do not rebuild a table for Y).
- Interval $[\mu - \sigma, \mu + \sigma]$; add the probabilities of the values inside it.

ANSWERS AT A GLANCEa) $E(X) = 6.3$, $\text{Var} = 8.61$, $\text{SD} = 2.934$ b) $E(Y) = 9.6$, $\text{Var}(Y) = 34.44$, $\text{SD}(Y) = 5.869$ c) Interval $[3.37, 9.23] \Rightarrow$ values 5, 8 $\Rightarrow 0.7$

STEP-BY-STEP SOLUTION

STEP part a Expectation table

x	p	xp	x^2p
2	0.2	0.4	0.8
5	0.3	1.5	7.5
8	0.4	3.2	25.6
12	0.1	1.2	14.4
Sum	1	6.3	48.3

$$E(X) = 6.3; \text{Var}(X) = 48.3 - 6.3^2 = 48.3 - 39.69 = 8.61; \text{SD}(X) = 2.934$$

STEP part b Linear transformation

$$E(Y) = 2(6.3) - 3 = 9.6$$

$$\text{Var}(Y) = 2^2 \times 8.61 = 34.44 \text{ (the } -3 \text{ does not change the spread)}$$

$$\text{SD}(Y) = 2 \times 2.934 = 5.869$$

9.6, 34.44, 5.869

STEP part c Within one SD

$$\mu - \sigma = 6.3 - 2.934 = 3.366; \mu + \sigma = 9.234$$

Values inside $[3.366, 9.234]$: 5 and 8

$$P = 0.3 + 0.4 = 0.7$$

0.7 — 70 % of the probability sits within one SD of the mean (for a normal it would be about 68 %; the discrete gaps make it slightly different).

WHERE MARKS ARE LOST

Writing $\text{Var}(2X - 3) = 2 \text{Var}(X) - 3 \cdot$ using $\text{Var}(Y) = 4\text{Var}(X) - 3 \cdot$ including 2 or 12 in the one-SD interval without checking the endpoints.

Q 5.4 Expected gain · fair game · entry fee · variance

4 marks

A stall runs the game: roll one fair die; win ₹6 if a 6 shows, win ₹2 if a 2 or a 4 shows, lose ₹3 otherwise. Let X be the player's gain.

- a) Write the distribution of X . [1]
- b) Find $E(X)$. Is the game fair? Who does it favour? [1.5]
- c) What entry fee would make the game fair? [0.5]
- d) Find $\text{Var}(X)$. [1]

WHAT THIS QUESTION TESTS

Building a pmf from a rule, expected value as long-run average gain, and the definition of a fair game.

HOW TO ATTEMPT

- Group the die faces by payout: $\{6\}$, $\{2, 4\}$, $\{1, 3, 5\}$.
- $E(X) = \sum x p(x)$; fair $\Leftrightarrow 0$.
- An entry fee equal to $E(X)$ makes the net expectation 0.

ANSWERS AT A GLANCE

a) 6 w.p. $1/6$, 2 w.p. $2/6$, -3 w.p. $3/6$ b) $E(X) = 1/6 = 0.1667 \Rightarrow$ not fair, favours the player c) ₹0.17 d) 11.806

STEP-BY-STEP SOLUTION**STEP** part a **Distribution**

Outcome	6	2 or 4	1, 3 or 5
x (₹)	6	2	-3
$p(x)$	$1/6$	$2/6$	$3/6$

STEP part b **Expected gain**

$$E(X) = 6 \cdot \frac{1}{6} + 2 \cdot \frac{2}{6} + (-3) \cdot \frac{3}{6} = 1 + 0.6667 - 1.5 = 0.1667$$

$E(X) = +0.17$ — that is, +₹0.17 per play — not fair; it favours the player (the stall loses on average).

STEP part c **Entry fee**

Charging an entry fee of ₹ $1/6 \approx ₹0.17$ makes the expected net gain zero, i.e. a fair game.

STEP part d **Variance**

$$E(X^2) = 36 \cdot \frac{1}{6} + 4 \cdot \frac{2}{6} + 9 \cdot \frac{3}{6} = 6 + 1.3333 + 4.5 = 11.8333$$

$$\text{Var}(X) = 11.8333 - 0.1667^2 = 11.8333 - 0.0278 = 11.806$$

$\text{Var} \approx 11.81$ (SD ≈ 3.44) — large spread relative to a tiny mean gain.

WHERE MARKS ARE LOST

Treating each die face as a separate value but giving the loss probability $1/6$ instead of $3/6$ · squaring -3 as -9 in $E(X^2)$ · calling the game “fair” because gains and losses “roughly cancel”.

Q 5.5 Sampling without replacement · full distribution · mean

4 marks

A lot of 8 components contains 2 defective ones. Three components are drawn at random without replacement. Let X be the number of defective components in the sample.

- a) Find the probability distribution of X . [2]
 b) Find $P(X \geq 1)$ and $F(1)$. [1]
 c) Find $E(X)$. [1]

WHAT THIS QUESTION TESTS

Counting with combinations for each value of X , checking the probabilities sum to 1, then the usual expectation.

HOW TO ATTEMPT

- Total samples $\binom{8}{3} = 56$.
- $P(X = k) = \binom{2}{k} \binom{6}{3-k} / 56$ for $k = 0, 1, 2$ (3 is impossible: only 2 defectives exist).
- Sum to check; then the mean.

ANSWERS AT A GLANCE

a) $P(0) = 20/56$, $P(1) = 30/56$, $P(2) = 6/56$ b) $P(X \geq 1) = 36/56 = 0.643$, $F(1) = 50/56 = 0.893$ c) $E(X) = 0.75$

STEP-BY-STEP SOLUTION

STEP **part a** **Combinations**

$$\binom{8}{3} = 56$$

$$P(X = 0) = \frac{\binom{2}{0}\binom{6}{3}}{56} = \frac{1 \times 20}{56} = \frac{20}{56} = 0.3571$$

$$P(X = 1) = \frac{\binom{2}{1}\binom{6}{2}}{56} = \frac{2 \times 15}{56} = \frac{30}{56} = 0.5357$$

$$P(X = 2) = \frac{\binom{2}{2}\binom{6}{1}}{56} = \frac{1 \times 6}{56} = \frac{6}{56} = 0.1071$$

$$\text{Check: } 20 + 30 + 6 = 56 \checkmark$$

x	0	1	2
$P(X = x)$	5/14	15/28	3/28

STEP **part b** **Tail and CDF**

$$P(X \geq 1) = 1 - 20/56 = 36/56 = 0.6429$$

$$F(1) = P(X \leq 1) = 50/56 = 0.8929$$

STEP **part c** **Mean**

$$E(X) = 0 \cdot \frac{20}{56} + 1 \cdot \frac{30}{56} + 2 \cdot \frac{6}{56} = \frac{42}{56} = 0.75$$

$$(\text{shortcut check: } n \cdot K/N = 3 \times 2/8 = 0.75 \checkmark)$$

0.75 defective components on average

WHERE MARKS ARE LOST

Using the binomial with $p = 2/8$ (draws are not independent without replacement) · including $X = 3$ with a non-zero probability · forgetting $\binom{6}{2} = 15$ in the $X = 1$ term.

Q 5.6 Joint pmf table · marginals · conditionals · independence · covariance

6 marks

A study of students recorded weekly study hours X (1, 2 or 3) and quiz score Y (10, 20 or 30). The joint pmf is:

$X \backslash Y$	10	20	30
1	0.10	0.15	0.05
2	0.05	0.20	0.15
3	0.02	0.08	0.20

- Find the marginal distributions of X and Y . [1]
- Find $P(X \geq 2, Y \leq 20)$. [1]
- Find $P(Y > 10 \mid X = 2)$ and $P(X = 1 \mid Y = 30)$. [1.5]
- Are X and Y independent? Justify. [1]
- Find $E(X)$, $E(Y)$, $E(XY)$ and $\text{Cov}(X, Y)$. [1.5]

WHAT THIS QUESTION TESTS

Every standard operation on a joint table. This is the most common joint-distribution question shape.

HOW TO ATTEMPT

- Add rows and columns; write them in the margins; check the grand total is 1.
- (b) is a block of cells. (c) divides a part of a row (or column) by that row's (column's) total.
- Independence: test one cell, e.g. (1, 10).
- $E(XY)$: multiply each cell by $x \cdot y$ and add.

ANSWERS AT A GLANCE

a) $f_X = (0.30, 0.40, 0.30)$, $f_Y = (0.17, 0.43, 0.40)$ b) 0.35 c) 0.875 and 0.125 d) No ($0.10 \neq 0.051$) e) 2.0, 22.3, 46.9, 2.3

STEP-BY-STEP SOLUTION**STEP** part a **Marginals**

$X \backslash Y$	10	20	30	$f_X(x)$
1	0.10	0.15	0.05	0.30
2	0.05	0.20	0.15	0.40
3	0.02	0.08	0.20	0.30
$f_Y(y)$	0.17	0.43	0.40	1.00

STEP part b **Block of cells**

Rows $X = 2, 3$; columns $Y = 10, 20$: $0.05 + 0.20 + 0.02 + 0.08 = 0.35$

0.35

STEP part c **Conditionals**

$$P(Y > 10 \mid X = 2) = \frac{0.20 + 0.15}{0.40} = \frac{0.35}{0.40} = 0.875$$

$$P(X = 1 \mid Y = 30) = \frac{0.05}{0.40} = 0.125$$

0.875 and 0.125

STEP part d **Independence**

Cell (1, 10): $f(1, 10) = 0.10$; $f_X(1)f_Y(10) = 0.30 \times 0.17 = 0.051 \neq 0.10$

Not independent — more study hours shift probability toward higher scores.

STEP part e **Expectations and covariance**

$$E(X) = 1(0.30) + 2(0.40) + 3(0.30) = 2.0$$

$$E(Y) = 10(0.17) + 20(0.43) + 30(0.40) = 1.7 + 8.6 + 12 = 22.3$$

$$E(XY): \text{row 1: } 1(10 \cdot 0.10 + 20 \cdot 0.15 + 30 \cdot 0.05) = 5.5; \text{ row 2: } 2(0.5 + 4 + 4.5) = 18; \text{ row 3: } 3(0.2 + 1.6 + 6) = 23.4$$

$$E(XY) = 5.5 + 18 + 23.4 = 46.9$$

$$\text{Cov}(X, Y) = 46.9 - 2.0 \times 22.3 = 46.9 - 44.6 = 2.3$$

$E(X) = 2$, $E(Y) = 22.3$, $E(XY) = 46.9$, $\text{Cov} = 2.3 > 0$ (positive association, consistent with (d))

WHERE MARKS ARE LOST

Conditioning on the wrong margin (dividing by $0.40 = f_Y(30)$ in the first conditional) · checking only one cell that happens to work and declaring independence · computing $E(XY)$ as $E(X)E(Y)$ (only valid if independent).

Q 5.7 Joint pmf by formula · constant · marginals · conditional · independence

5 marks

Two discrete random variables have joint pmf $f(x, y) = c(x + y)$ for $x \in \{1, 2, 3\}$, $y \in \{1, 2\}$.

- a) Find c . [1]
 b) Find the marginal pmfs of X and Y . [1.5]
 c) Find $P(X = 2, Y \geq 1)$ and $P(Y = 2 \mid X \geq 2)$. [1.5]
 d) Are X and Y independent? [1]

WHAT THIS QUESTION TESTS

Turning a formula into a table, then the usual table operations; a conditional on an *event* ($X \geq 2$), not just a single value.

HOW TO ATTEMPT

1. Tabulate $x + y$ for the six cells; sum = 21 $\Rightarrow c = 1/21$.
2. Row and column sums.
3. Conditional on $X \geq 2$: cells in rows 2 and 3 with $Y = 2$, divided by all cells in rows 2 and 3.

ANSWERS AT A GLANCE

a) $c = 1/21$ b) $f_X = (5, 7, 9)/21$, $f_Y = (9, 12)/21$ c) $7/21 = 1/3$ and $9/16$ d) No

STEP-BY-STEP SOLUTION**STEP** part a Table of $x + y$

$x \backslash y$	1	2	row sum
1	$2c$	$3c$	$5c$
2	$3c$	$4c$	$7c$
3	$4c$	$5c$	$9c$
col sum	$9c$	$12c$	$21c$

$$21c = 1 \Rightarrow c = 1/21$$

STEP part b Marginals

$$f_X(1) = 5/21, f_X(2) = 7/21, f_X(3) = 9/21$$

$$f_Y(1) = 9/21, f_Y(2) = 12/21$$

STEP part c Joint and conditional on an event

$$P(X = 2, Y \geq 1) = f(2, 1) + f(2, 2) = 7/21 = 1/3 \text{ (the whole row, since } Y \geq 1 \text{ always)}$$

$$P(X \geq 2) = (7 + 9)/21 = 16/21; P(Y = 2, X \geq 2) = (4 + 5)/21 = 9/21$$

$$P(Y = 2 \mid X \geq 2) = \frac{9/21}{16/21} = \frac{9}{16} = 0.5625$$

1/3 and 9/16

STEP part d Independence

$$f(1, 1) = 2/21 = 0.0952; f_X(1)f_Y(1) = \frac{5}{21} \cdot \frac{9}{21} = 45/441 = 0.1020$$

Not equal $\Rightarrow$ not independent (a sum $x + y$ can never factor into a product $g(x)h(y)$).

WHERE MARKS ARE LOST

Summing $x + y$ over the wrong ranges (e.g. treating y as 0, 1, 2) · in the conditional on $X \geq 2$, dividing by $f_X(2)$ only · leaving answers as multiples of c after c is known.

Q 5.8 Joint distribution from an experiment · marginals · $E(XY)$

5 marks

A fair coin is tossed three times. Let X = number of heads, and $Y = 1$ if the first toss is a head and $Y = 0$ otherwise.

- a) List the 8 outcomes with their (X, Y) values and write the joint probability table. [2]
- b) Find the marginal pmfs of X and Y . [1]
- c) Find $P(X \geq 2 | Y = 1)$. [1]
- d) Find $E(XY)$ and $\text{Cov}(X, Y)$. [1]

WHAT THIS QUESTION TESTS

Constructing a joint pmf from the sample space rather than being handed a table; then marginals, a conditional, and covariance.

HOW TO ATTEMPT

1. Enumerate HHH ... TTT (each $1/8$); record X and Y .
2. Group outcomes with the same (X, Y) .
3. Margins; conditional divides by $P(Y = 1) = 1/2$; $E(XY)$ only needs the $Y = 1$ column.

ANSWERS AT A GLANCE

- a) Non-zero cells: (0,0) $1/8$, (1,0) $2/8$, (1,1) $1/8$, (2,0) $1/8$, (2,1) $2/8$, (3,1) $1/8$
 b) $f_X = (1, 3, 3, 1)/8$; $f_Y = (1/2, 1/2)$ c) $3/4$ d) $E(XY) = 1$, $\text{Cov} = 0.25$

STEP-BY-STEP SOLUTION**STEP** part a Enumerate

Outcome	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
X	3	2	2	1	2	1	1	0
Y	1	1	1	1	0	0	0	0

$X \backslash Y$	0	1	f_X
0	$1/8$	0	$1/8$
1	$2/8$	$1/8$	$3/8$
2	$1/8$	$2/8$	$3/8$
3	0	$1/8$	$1/8$
f_Y	$4/8$	$4/8$	1

STEP part b Marginals

$f_X = (1/8, 3/8, 3/8, 1/8)$ for $x = 0, 1, 2, 3$ (the Binomial(3, $1/2$) shape, as it should be)
 $f_Y(0) = f_Y(1) = 1/2$

STEP part c Conditional

$$P(X \geq 2 | Y = 1) = \frac{2/8 + 1/8}{4/8} = \frac{3}{4}$$

0.75 — if the first toss is a head, only one more head is needed among two tosses.

STEP **part d** **$E(XY)$ and covariance**

Only cells with $Y = 1$ contribute: $E(XY) = 1 \cdot \frac{1}{8} + 2 \cdot \frac{2}{8} + 3 \cdot \frac{1}{8} = \frac{8}{8} = 1$

$E(X) = 1.5$, $E(Y) = 0.5$

$\text{Cov}(X, Y) = 1 - 1.5 \times 0.5 = 0.25$

$E(XY) = 1$, $\text{Cov}(X, Y) = 0.25$ (positive: a head on toss 1 raises the head count)

WHERE MARKS ARE LOST

Assigning probabilities other than $1/8$ to outcomes · forgetting the zero cells (0,1) and (3,0) — they must appear as 0 in the table · dividing by $1/8$ instead of by $P(Y = 1) = 1/2$ in (c).

UNIT 6

Continuous Random Variables

pdf and its two conditions · finding the constant by integration · probabilities as areas · expectation and variance by integration · CDF $\leftrightarrow$ pdf · joint pdf · marginal densities · independence · $E(XY)$

What this unit is about

When a variable can take any value in an interval (time, length, a rating between 0 and 2), single values have probability zero and we work with a **probability density function** $f(x)$: probabilities are **areas under the curve**. The two conditions are $f(x) \geq 0$ and total area 1 — the second is how the unknown constant is found. The **CDF** $F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$ accumulates area from the left; differentiating it gives the pdf back. Every discrete sum in Unit 5 becomes an integral here: $E(X) = \int x f(x) dx$, $E(X^2) = \int x^2 f(x) dx$, $\text{Var} = E(X^2) - [E(X)]^2$.

For **two continuous variables** the joint pdf $f(x, y)$ integrates to 1 over the region; marginal densities are obtained by integrating out the other variable; independence means $f(x, y) = f_X(x)f_Y(y)$ — which is easy to spot: if $f(x, y)$ is a product of a function of x and a function of y on a rectangle, they are independent; a sum like $x + 2y$ never factors.

How to attempt questions from this unit

- Constant:** integrate f over its support piece by piece (piecewise pdfs are common — one integral per piece), set the total to 1, solve.
- Probabilities:** $P(a < X < b) = \int_a^b f(x) dx$. Split the integral where the formula changes. If a CDF is given, use $F(b) - F(a)$ instead — no integration needed.
- Moments:** $E(X) = \int x f$, $E(X^2) = \int x^2 f$; write the antiderivative, substitute limits on one line each. Then variance and SD.
- CDF $\leftrightarrow$ pdf:** pdf $\rightarrow$ CDF by integrating from the lower limit (and continuing with 1 after the support); CDF $\rightarrow$ pdf by differentiating each piece. Check $F(\text{upper limit}) = 1$ to find a constant in a CDF.
- Joint pdf:** double integral for the constant; marginal $f_X(x) = \int f(x, y) dy$ (integrate the *other* variable over its range); $P(X < a, Y > b) = \int_b^a \int_a^b f dx dy$; independence by factoring; $E(XY) = \iint xy f$.
- Interpretation:** quote the mean as “average lifetime”, a tail probability as “fraction that survive beyond ...”, and the SD as the spread.

Formula card

Valid pdf: $f \geq 0$, $\int_{-\infty}^{\infty} f(x) dx = 1$

$$F(x) = \int_{-\infty}^x f(t) dt, f(x) = F'(x)$$

$$E(X) = \int x f(x) dx, E(X^2) = \int x^2 f(x) dx$$

$$\text{Joint: } \iint f(x, y) dx dy = 1$$

$$\text{Independent} \Leftrightarrow f(x, y) = f_X(x)f_Y(y) \text{ everywhere}$$

$$\text{Median } m: F(m) = 0.5$$

$$P(a \leq X \leq b) = \int_a^b f(x) dx = F(b) - F(a)$$

$$P(X = a) = 0; P(X > a) = 1 - F(a)$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\text{Marginals: } f_X(x) = \int f(x, y) dy, f_Y(y) = \int f(x, y) dx$$

$$E(XY) = \iint xy f(x, y) dx dy; \text{ if independent } E(XY) = E(X)E(Y)$$

$$\text{Useful: } \int_0^a x^n dx = \frac{a^{n+1}}{n+1}$$

Question map

Q	Shape of the question	Marks
6.1	Linear pdf on $[0, 4]$: constant, tail probability, mean, variance, CDF, median, interpretation	5
6.2	Piecewise pdf: constant, probability across the join, mean, variance	5
6.3	CDF given: constant, interval probability, derive the pdf, mean	5
6.4	Quadratic pdf on $[0, 1]$: constant, probability, mean, variance, SD	4
6.5	Joint pdf that factors: constant, marginals, independence, joint probability, $E(X)$, $E(Y)$, $E(XY)$	5
6.6	Joint pdf that does not factor: constant, marginals, independence, a rectangle probability, $E(X)$	5

Q 6.1 Linear pdf · tail probability · mean · variance · CDF · median

5 marks

The time X (hours) a rechargeable lamp runs before needing a recharge has pdf $f(x) = kx$ for $0 \leq x \leq 4$, and 0 otherwise.

- a) Find k . [1]
 b) Probability that the lamp runs more than 3 hours. [1]
 c) Find $E(X)$, $\text{Var}(X)$ and the SD. [2]
 d) Write the CDF $F(x)$ and find the median running time. [1]

WHAT THIS QUESTION TESTS

The complete single-variable workflow on the simplest possible pdf, including the CDF and median.

HOW TO ATTEMPT

- $\int_0^4 kx \, dx = 8k = 1$.
- Tail = $\int_3^4$.
- $\int x \cdot x/8$ and $\int x^2 \cdot x/8$.
- $F(x) = x^2/16$; set equal to 0.5.

ANSWERS AT A GLANCE

a) $k = 1/8$ b) $7/16 = 0.4375$ c) $E(X) = 8/3 = 2.667$, $\text{Var} = 8/9 = 0.889$, SD 0.943 d) $F(x) = x^2/16$, median $\sqrt{8} = 2.83$ h

STEP-BY-STEP SOLUTION

STEP part a Area = 1

$$\int_0^4 kx \, dx = k \left[\frac{x^2}{2} \right]_0^4 = 8k = 1 \Rightarrow k = \frac{1}{8}$$

$f(x) = x/8$ on $[0, 4]$

STEP part b Tail probability

$$P(X > 3) = \int_3^4 \frac{x}{8} \, dx = \frac{1}{16} [x^2]_3^4 = \frac{16 - 9}{16} = \frac{7}{16} = 0.4375$$

0.4375

STEP **part c** Mean and variance

$$E(X) = \int_0^4 x \cdot \frac{x}{8} dx = \frac{1}{8} \left[\frac{x^3}{3} \right]_0^4 = \frac{64}{24} = \frac{8}{3} = 2.667$$

$$E(X^2) = \int_0^4 \frac{x^3}{8} dx = \frac{1}{8} \left[\frac{x^4}{4} \right]_0^4 = \frac{256}{32} = 8$$

$$\text{Var}(X) = 8 - \left(\frac{8}{3} \right)^2 = 8 - \frac{64}{9} = \frac{8}{9} = 0.889; \text{SD} = 0.943$$

Mean 2.67 h, variance 0.889 h², SD 0.943 h

STEP **part d** CDF and median

$$F(x) = \int_0^x \frac{t}{8} dt = \frac{x^2}{16} \text{ for } 0 \leq x \leq 4 \text{ (0 before, 1 after)}$$

$$F(m) = 0.5 \Rightarrow m^2 = 8 \Rightarrow m = 2.83$$

Median 2.83 h — half the lamps last less than 2.83 h; note the mean (2.67) is below the median because the density rises to the right (left-skewed).

WHERE MARKS ARE LOST

Forgetting to multiply by x in $E(X)$ (integrating the pdf again) · computing $P(X > 3)$ as $f(3)$ · writing the CDF without its 0 / 1 pieces outside the support.

Q 6.2 Piecewise pdf · constant · probability across the join · mean · variance

5 marks

The response time X (seconds) of a service has pdf

$$f(x) = \begin{cases} c, & 0 \leq x \leq 1 \\ \frac{c(3-x)}{2}, & 1 < x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

- a) Find c . [1.5]
 b) Find $P(0.5 < X < 2)$. [1.5]
 c) Find $E(X)$ and $\text{Var}(X)$. [2]

WHAT THIS QUESTION TESTS

Handling a pdf with two formulas: one integral per piece, every time.

HOW TO ATTEMPT

1. Area of piece 1 (a rectangle) + area of piece 2 (a triangle) = 1.
2. Split (0.5, 2) at $x = 1$.
3. Same split for $E(X)$ and $E(X^2)$.

ANSWERS AT A GLANCE

a) $c = 1/2$ b) 0.625 c) $E(X) = 1.0833$, $E(X^2) = 1.6667$, $\text{Var} = 0.4931$

STEP-BY-STEP SOLUTION

STEP **part a** Two pieces

Piece 1: $\int_0^1 c \, dx = c$

Piece 2: $\int_1^3 \frac{c(3-x)}{2} \, dx = \frac{c}{2} \left[3x - \frac{x^2}{2} \right]_1^3 = \frac{c}{2} [(9 - 4.5) - (3 - 0.5)] = \frac{c}{2}(2) = c$

$c + c = 1 \Rightarrow c = \frac{1}{2}$

$c = 1/2$

STEP **part b** Split at $x = 1$

$P(0.5 < X < 1) = \int_{0.5}^1 \frac{1}{2} \, dx = 0.25$

$P(1 < X < 2) = \int_1^2 \frac{3-x}{4} \, dx = \frac{1}{4} \left[3x - \frac{x^2}{2} \right]_1^2 = \frac{1}{4} [(6 - 2) - (3 - 0.5)] = \frac{1}{4}(1.5) = 0.375$

Total = $0.25 + 0.375 = 0.625$

0.625

STEP **part c** Mean and variance

$$E(X) = \int_0^1 \frac{x}{2} \, dx + \int_1^3 \frac{x(3-x)}{4} \, dx = \frac{1}{4} + \frac{1}{4} \left[\frac{3x^2}{2} - \frac{x^3}{3} \right]_1^3$$

$$= 0.25 + \frac{1}{4} [(13.5 - 9) - (1.5 - 0.3333)] = 0.25 + \frac{1}{4}(3.3333) = 0.25 + 0.8333 = 1.0833$$

$$E(X^2) = \int_0^1 \frac{x^2}{2} \, dx + \int_1^3 \frac{x^2(3-x)}{4} \, dx = \frac{1}{6} + \frac{1}{4} \left[x^3 - \frac{x^4}{4} \right]_1^3$$

$$= 0.1667 + \frac{1}{4} [(27 - 20.25) - (1 - 0.25)] = 0.1667 + \frac{1}{4}(6) = 1.6667$$

$\text{Var}(X) = 1.6667 - 1.0833^2 = 1.6667 - 1.1736 = 0.4931$

Mean 1.083 s, variance 0.493 s² (SD 0.702 s)

WHERE MARKS ARE LOST

Integrating the second piece from 0 instead of from 1 · dropping the factor $\frac{1}{2}$ in $c(3-x)/2$ after finding c · evaluating $[3x - x^2/2]$ at only the upper limit.

Q 6.3 CDF given · constant · interval probability · pdf by differentiation · mean**5 marks**

The normalised load index X of a server (0 = idle, 2 = overloaded) has CDF

$$F(x) = \begin{cases} 0, & x < 0 \\ kx^2, & 0 \leq x \leq 1 \\ k(2x - 1), & 1 < x \leq 2 \\ 1, & x > 2 \end{cases}$$

- Find k . [1.5]
- Find the probability that the server is in the warning zone $0.5 \leq X \leq 1.5$, and $P(X > 1.5)$. [1.5]
- Derive the pdf $f(x)$. [1]
- Find $E(X)$. [1]

WHAT THIS QUESTION TESTS

Working from a CDF: the constant comes from $F(2) = 1$ (not from an integral), probabilities come from subtraction, and the pdf from differentiation.

HOW TO ATTEMPT

1. F must reach 1 at the top of the support: $F(2) = k(4 - 1) = 3k = 1$. Also check continuity at $x = 1$.
2. $P(a \leq X \leq b) = F(b) - F(a)$.
3. Differentiate each piece.
4. $E(X) = \int xf$ with the pdf from (c).

ANSWERS AT A GLANCE

a) $k = 1/3$ b) 0.5833 and 0.3333 c) $f(x) = 2x/3$ on $[0,1]$, $2/3$ on $(1,2]$, 0 otherwise d) $E(X) = 11/9 = 1.222$

STEP-BY-STEP SOLUTION**STEP** part a **Constant from $F(2) = 1$**

$$F(2) = k(2 \cdot 2 - 1) = 3k = 1 \Rightarrow k = \frac{1}{3}$$

Continuity at 1: $k(1)^2 = k$ and $k(2 - 1) = k$ ✓ (both pieces agree)

$$k = 1/3$$

STEP part b **Subtract CDF values**

$$F(1.5) = \frac{1}{3}(3 - 1) = \frac{2}{3}; F(0.5) = \frac{1}{3}(0.25) = \frac{1}{12}$$

$$P(0.5 \leq X \leq 1.5) = \frac{2}{3} - \frac{1}{12} = \frac{7}{12} = 0.5833$$

$$P(X > 1.5) = 1 - F(1.5) = \frac{1}{3} = 0.3333$$

0.5833 and 0.3333

STEP part c **Differentiate**

$$0 \leq x \leq 1: f(x) = \frac{d}{dx} \left(\frac{x^2}{3} \right) = \frac{2x}{3}$$

$$1 < x \leq 2: f(x) = \frac{d}{dx} \left(\frac{2x - 1}{3} \right) = \frac{2}{3}$$

$$\text{Check area: } \int_0^1 \frac{2x}{3} dx + \int_1^2 \frac{2}{3} dx = \frac{1}{3} + \frac{2}{3} = 1 \quad \checkmark$$

$f(x) = 2x/3$ on $[0,1]$, $2/3$ on $(1,2]$, 0 elsewhere

STEP part d **Mean**

$$E(X) = \int_0^1 x \cdot \frac{2x}{3} dx + \int_1^2 x \cdot \frac{2}{3} dx = \frac{2}{9} + \frac{2}{3} \left[\frac{x^2}{2} \right]_1^2 = \frac{2}{9} + \frac{2}{3} \cdot \frac{3}{2} = \frac{2}{9} + 1 = \frac{11}{9}$$

$$E(X) = 1.222$$

WHERE MARKS ARE LOST

Trying to find k by integrating the CDF (a CDF is not a density) · $P(X > 1.5)$ as $F(1.5)$ · forgetting that the pdf is 0 outside $[0, 2]$.

Q 6.4 Quadratic pdf · probability · mean · variance

4 marks

A recommendation score $X \in [0, 1]$ has pdf $f(x) = k(1 - x^2)$, $0 \leq x \leq 1$.

- a) Find k . [1]
- b) Find $P(X < 0.5)$. [1]
- c) Find $E(X)$, $\text{Var}(X)$ and the SD. [2]

WHAT THIS QUESTION TESTS

Polynomial integration done cleanly, with the power rule applied line by line.

HOW TO ATTEMPT

1. $\int_0^1 (1 - x^2) = 1 - \frac{1}{3} = \frac{2}{3}$.
2. Same antiderivative up to 0.5.
3. $E(X) = k \int (x - x^3)$, $E(X^2) = k \int (x^2 - x^4)$.

ANSWERS AT A GLANCE

a) $k = 3/2$ b) 0.6875 c) $E(X) = 0.375$, $E(X^2) = 0.2$, Var 0.0594, SD 0.244

STEP-BY-STEP SOLUTION**STEP** part a **Constant**

$$\int_0^1 k(1 - x^2)dx = k \left[x - \frac{x^3}{3} \right]_0^1 = k \cdot \frac{2}{3} = 1 \Rightarrow k = \frac{3}{2}$$

STEP part b **Probability**

$$P(X < 0.5) = \frac{3}{2} \left[x - \frac{x^3}{3} \right]_0^{0.5} = \frac{3}{2} \left(0.5 - \frac{0.125}{3} \right) = \frac{3}{2}(0.45833) = 0.6875$$

0.6875**STEP** part c **Moments**

$$E(X) = \frac{3}{2} \int_0^1 (x - x^3)dx = \frac{3}{2} \left(\frac{1}{2} - \frac{1}{4} \right) = \frac{3}{2} \cdot \frac{1}{4} = 0.375$$

$$E(X^2) = \frac{3}{2} \int_0^1 (x^2 - x^4)dx = \frac{3}{2} \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{3}{2} \cdot \frac{2}{15} = 0.2$$

$$\text{Var}(X) = 0.2 - 0.375^2 = 0.2 - 0.140625 = 0.059375; \text{SD} = 0.2437$$

Mean 0.375, Var 0.0594, SD 0.244**WHERE MARKS ARE LOST**

Integrating $(1 - x^2)$ as $x - x^3$ (missing the $\div 3$) · using $k = 2/3$ (the reciprocal) · computing the variance as $E(X^2) - E(X)$.

Q 6.5 Joint pdf that factors · marginals · independence · $E(XY)$ **5 marks**

Let X and Y have joint pdf $f(x, y) = kxy^2$ for $0 \leq x \leq 2$, $0 \leq y \leq 1$, and 0 otherwise.

- a) Find k . [1]
- b) Find the marginal densities $f_X(x)$ and $f_Y(y)$. [1.5]
- c) Are X and Y independent? Justify. [0.5]
- d) Compute $P(X < 1, Y > 0.5)$. [1]
- e) Compute $E(X)$, $E(Y)$ and $E(XY)$. [1]

WHAT THIS QUESTION TESTS

The double-integral workflow, and the shortcut that a product pdf on a rectangle means independence, which in turn lets probabilities and $E(XY)$ factor.

HOW TO ATTEMPT

1. $\iint kxy^2 = k \int_0^2 x dx \int_0^1 y^2 dy = k \cdot 2 \cdot \frac{1}{3}$.
2. Marginal of X : integrate over y from 0 to 1; marginal of Y : integrate over x from 0 to 2.
3. Check $f_X f_Y = f$. Then (d) and (e) factor.

ANSWERS AT A GLANCE

a) $k = 3/2$ b) $f_X(x) = x/2$ on $[0, 2]$, $f_Y(y) = 3y^2$ on $[0, 1]$ c) Yes d) 0.21875 e) $E(X) = 4/3$, $E(Y) = 3/4$, $E(XY) = 1$

STEP-BY-STEP SOLUTION**STEP** part a **Constant**

$$\int_0^1 \int_0^2 kxy^2 dx dy = k \left[\frac{x^2}{2} \right]_0^2 \left[\frac{y^3}{3} \right]_0^1 = k \cdot 2 \cdot \frac{1}{3} = \frac{2k}{3} = 1 \Rightarrow k = \frac{3}{2}$$

STEP part b **Marginals**

$$f_X(x) = \int_0^1 \frac{3}{2}xy^2 dy = \frac{3}{2}x \left[\frac{y^3}{3} \right]_0^1 = \frac{x}{2}, 0 \leq x \leq 2$$

$$f_Y(y) = \int_0^2 \frac{3}{2}xy^2 dx = \frac{3}{2}y^2 \left[\frac{x^2}{2} \right]_0^2 = 3y^2, 0 \leq y \leq 1$$

$$\text{Checks: } \int_0^2 x/2 dx = 1 \checkmark, \int_0^1 3y^2 dy = 1 \checkmark$$

STEP part c **Independence**

$$f_X(x)f_Y(y) = \frac{x}{2} \cdot 3y^2 = \frac{3}{2}xy^2 = f(x, y) \text{ for every } (x, y) \text{ in the rectangle}$$

Independent.

STEP part d **Rectangle probability (factor because independent)**

$$P(X < 1) = \int_0^1 \frac{x}{2} dx = \frac{1}{4}; P(Y > 0.5) = \int_{0.5}^1 3y^2 dy = 1 - 0.125 = 0.875$$

$$P(X < 1, Y > 0.5) = 0.25 \times 0.875 = 0.21875$$

0.21875

STEP part e **Expectations**

$$E(X) = \int_0^2 x \cdot \frac{x}{2} dx = \frac{8}{6} = \frac{4}{3}; E(Y) = \int_0^1 y \cdot 3y^2 dy = \frac{3}{4}$$

$$\text{Independent} \Rightarrow E(XY) = E(X)E(Y) = \frac{4}{3} \cdot \frac{3}{4} = 1 \text{ (direct: } \iint xy \cdot \frac{3}{2}xy^2 = \frac{3}{2} \cdot \frac{8}{3} \cdot \frac{1}{4} = 1 \checkmark)$$

4/3, 3/4, 1

WHERE MARKS ARE LOST

Integrating f_X over x instead of over y · limits swapped (x goes to 2, y goes to 1) · claiming $E(XY) = E(X)E(Y)$ without first establishing independence.

Q 6.6 Joint pdf that does not factor · marginals · dependence · rectangle probability

5 marks

Let $f(x, y) = k(x + 2y)$ for $0 \leq x \leq 1$, $0 \leq y \leq 1$, and 0 otherwise.

- a) Find k . [1]
- b) Find $f_X(x)$ and $f_Y(y)$. [1.5]
- c) Are X and Y independent? Justify. [1]
- d) Find $P(X < 0.5, Y < 0.5)$. [1]
- e) Find $E(X)$. [0.5]

WHAT THIS QUESTION TESTS

The non-separable case: marginals must be computed by integration, independence fails, and a rectangle probability needs the actual double integral.

HOW TO ATTEMPT

1. $\iint (x + 2y) = \frac{1}{2} + 1 = \frac{3}{2}$.
2. Integrate out y for f_X , out x for f_Y .
3. Compare product with f at any point, e.g. $(0, 0)$.
4. Inner integral in y , then outer in x .

ANSWERS AT A GLANCE

a) $k = 2/3$ b) $f_X(x) = \frac{2}{3}(x + 1)$, $f_Y(y) = \frac{1}{3}(1 + 4y)$ c) No d) 0.125 e) $5/9 = 0.556$

STEP-BY-STEP SOLUTION**STEP** part a **Constant**

$$\int_0^1 \int_0^1 k(x + 2y) dy dx = k \int_0^1 [xy + y^2]_0^1 dx = k \int_0^1 (x + 1) dx = k \left(\frac{1}{2} + 1 \right) = \frac{3k}{2} = 1 \Rightarrow k = \frac{2}{3}$$

STEP part b **Marginals**

$$f_X(x) = \int_0^1 \frac{2}{3}(x + 2y) dy = \frac{2}{3} [xy + y^2]_0^1 = \frac{2}{3}(x + 1), 0 \leq x \leq 1$$

$$f_Y(y) = \int_0^1 \frac{2}{3}(x + 2y) dx = \frac{2}{3} \left[\frac{x^2}{2} + 2xy \right]_0^1 = \frac{2}{3} \left(\frac{1}{2} + 2y \right) = \frac{1 + 4y}{3}, 0 \leq y \leq 1$$

$$\text{Checks: } \int_0^1 \frac{2}{3}(x + 1) dx = \frac{2}{3} \cdot \frac{3}{2} = 1 \checkmark; \int_0^1 \frac{1 + 4y}{3} dy = \frac{1 + 2}{3} = 1 \checkmark$$

STEP part c **Independence**

$$\text{At } (0, 0): f(0, 0) = 0 \text{ but } f_X(0)f_Y(0) = \frac{2}{3} \cdot \frac{1}{3} = \frac{2}{9} \neq 0$$

Not independent — a sum $x + 2y$ cannot be written as a product $g(x)h(y)$.

STEP part d **Rectangle probability**

$$\begin{aligned} P &= \int_0^{0.5} \int_0^{0.5} \frac{2}{3}(x + 2y) dy dx = \frac{2}{3} \int_0^{0.5} [xy + y^2]_0^{0.5} dx = \frac{2}{3} \int_0^{0.5} (0.5x + 0.25) dx \\ &= \frac{2}{3} [0.25x^2 + 0.25x]_0^{0.5} = \frac{2}{3} (0.0625 + 0.125) = \frac{2}{3} (0.1875) = 0.125 \end{aligned}$$

0.125

STEP part e **E(X) from the marginal**

$$E(X) = \int_0^1 x \cdot \frac{2}{3}(x + 1) dx = \frac{2}{3} \left(\frac{1}{3} + \frac{1}{2} \right) = \frac{2}{3} \cdot \frac{5}{6} = \frac{5}{9} = 0.556$$

0.556

WHERE MARKS ARE LOST

Multiplying $P(X < 0.5)P(Y < 0.5)$ in (d) (not independent!) · forgetting the factor k in the marginals · an inner integral evaluated with the wrong variable's limits.

UNIT 7

Standard Distributions

Bernoulli · Binomial · Poisson · Normal and the standard normal table · Poisson approximation to Binomial · Normal approximation to Binomial with continuity correction · recovering parameters

What this unit is about

Four named distributions cover almost every counting or measuring situation in the syllabus. **Bernoulli**: one trial, success/failure. **Binomial**: the number of successes in n independent trials with the same success probability p . **Poisson**: the number of events in a fixed interval when events happen at a steady average rate λ (calls per hour, flaws per metre). **Normal**: the bell curve for measurements, fully described by mean μ and SD σ , and always used through the standard score $z = (x - \mu)/\sigma$ and the table of $\Phi(z) = P(Z \leq z)$.

Two bridges connect them and are examined every time. When n is large and p small, Binomial $\approx$ Poisson with $\lambda = np$. When $np \geq 15$ and $n(1 - p) \geq 15$, Binomial $\approx$ Normal with $\mu = np$, $\sigma = \sqrt{np(1 - p)}$, and because you are replacing a staircase by a curve you must apply the **continuity correction** of ± 0.5 . The exam also likes the reverse direction: given the mean and variance, recover n and p ; given two normal probabilities, recover μ and σ .

How to attempt questions from this unit

- Identify the distribution first** and write its parameters: " $X \sim B(40, 0.03)$ ", " $X \sim \text{Poisson}(\lambda = 6)$ ", " $X \sim N(50, 8^2)$ ". One mark is usually reserved for this line.
- Poisson rate scaling**: λ is per *interval*. Rate 2 per hour $\Rightarrow \lambda = 6$ for 3 hours, $\lambda = 0.5$ for 15 minutes. Always rescale before substituting.
- Cumulative probabilities**: write the inequality, then the sum. "At least 2" = $1 - P(0) - P(1)$; "at most 2" = $P(0) + P(1) + P(2)$; "between 2 and 5 inclusive" = $P(2) + \dots + P(5)$; "more than 4" = $1 - P(X \leq 4)$. Tabulate the terms.
- Normal**: sketch, convert each limit to z (2 decimals), read $\Phi(z)$, then: $P(X < a) = \Phi(z_a)$; $P(X > a) = 1 - \Phi(z_a)$; $P(a < X < b) = \Phi(z_b) - \Phi(z_a)$; $\Phi(-z) = 1 - \Phi(z)$. For a cut-off, read the table backwards: find z for the given area, then $x = \mu + z\sigma$.
- Approximations**: state the condition you checked (n large, p small; or $np, nq \geq 15$), compute the parameters, and for the normal case apply ± 0.5 to every discrete limit before converting to z .
- Expected counts**: probability $\times$ number of items (e.g. $0.1096 \times 1000 \approx 110$ defective bolts). Round to a whole number.

Formula cardBernoulli(p): $E = p$, $\text{Var} = p(1 - p)$ Poisson: $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$, $E = \text{Var} = \lambda$ Binomial $\approx$ Poisson: n large, p small, $\lambda = np$ Continuity: $P(X \leq k) \rightarrow P(X < k + 0.5)$; $P(X \geq k) \rightarrow P(X > k - 0.5)$ Normal: $z = \frac{x - \mu}{\sigma}$, $\Phi(-z) = 1 - \Phi(z)$ Key z : $\Phi(1.28) = 0.90$, $\Phi(1.645) = 0.95$, $\Phi(1.96) = 0.975$, $\Phi(2.33) = 0.99$, $\Phi(2.576) = 0.995$ Given $E = np$ and $\text{Var} = npq$: $q = \text{Var}/E$, $p = 1 - q$, $n = E/p$ Binomial: $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$, $E = np$, $\text{Var} = np(1 - p)$ Poisson recursion: $P(k + 1) = P(k) \cdot \frac{\lambda}{k + 1}$ Binomial $\approx$ Normal: $np \geq 15$, $nq \geq 15$; $\mu = np$, $\sigma = \sqrt{npq}$ $P(X < k) \rightarrow P(X < k - 0.5)$; $P(X > k) \rightarrow P(X > k + 0.5)$ Empirical rule: 68 % within 1σ , 95 % within 2σ , 99.7 % within 3σ Quartiles of a normal: $\mu \pm 0.674\sigma$ Given $P(X = 0) = e^{-\lambda}$: $\lambda = -\ln P(0)$ **Question map**

Q	Shape of the question	Marks
7.1	Binomial: exact, at most, at least, mean and variance; when is a situation binomial?	5
7.2	Binomial from mean and variance: recover n and p , write the pmf, $E(X^2)$, impossible range; a second recover-the-variance part	5
7.3	Guessing on an MCQ: at least 4 correct, at most 1, mean and variance	4
7.4	Poisson with rate scaling: one hour, three hours, 15 minutes, two consecutive periods	5
7.5	Poisson approximation to Binomial with expected counts; compare with the exact binomial	5
7.6	Binomial vs Poisson for the same probability: compute both, compare, explain	3
7.7	Normal probabilities: below, between, above, expected count	5
7.8	Inverse normal: top-5 % cut-off, bottom-10 % cut-off, middle 50 %, verify a claim	5
7.9	Specification limits: proportion defective, expected count, process-capability comment	5
7.10	Normal approximation to Binomial with continuity correction	5
7.11	Recover μ and σ from two normal probabilities; recover λ from $P(X = 0)$	4

Q 7.1 Binomial · exact and cumulative · mean and variance · conditions

5 marks

Each memory card produced by a line is defective with probability 0.10, independently. A pack contains 8 cards. Let X be the number of defective cards in a pack.

- State the distribution of X and find $P(X = 2)$. [1.5]
- Find $P(X \leq 1)$ and $P(X \geq 2)$. [1.5]
- Find $E(X)$ and $\text{Var}(X)$. [1]
- Which of these can be modelled as binomial, and why? (i) The number of defective cards when 3 cards are drawn without replacement from a box of 20 containing 4 defectives. (ii) The number of customers, out of 50 independent callers, who reach a live agent, each with probability 0.7. [1]

WHAT THIS QUESTION TESTS

Binomial pmf arithmetic, cumulative sums via the complement, and the four binomial conditions (fixed n , independent trials, constant p , two outcomes).

HOW TO ATTEMPT

1. $X \sim B(8, 0.1)$.
2. Compute $P(0), P(1), P(2)$ in a small table; reuse them.
3. np, npq .
4. Without replacement $\Rightarrow p$ changes $\Rightarrow$ not binomial.

ANSWERS AT A GLANCE

a) $B(8, 0.1)$; 0.1488 b) 0.8131 and 0.1869 c) 0.8 and 0.72 d) (i) no, (ii) yes

STEP-BY-STEP SOLUTION**STEP** **part a** **Distribution and $P(X = 2)$**

$$X \sim B(n = 8, p = 0.1), q = 0.9$$

$$P(X = 2) = \binom{8}{2}(0.1)^2(0.9)^6 = 28 \times 0.01 \times 0.531441 = 0.1488$$

0.1488

STEP **part b** **Cumulative**

k	$\binom{8}{k}$	$P(X = k)$
0	1	$0.9^8 = 0.4305$
1	8	$8(0.1)(0.9)^7 = 0.3826$

$$P(X \leq 1) = 0.4305 + 0.3826 = 0.8131$$

$$P(X \geq 2) = 1 - 0.8131 = 0.1869$$

0.8131 and 0.1869

STEP **part c** **Mean and variance**

$$E(X) = np = 0.8; \text{Var}(X) = npq = 8(0.1)(0.9) = 0.72$$

STEP **part d** **Binomial or not**

(i) Without replacement the defect probability changes from draw to draw (4/20, then 3/19 or 4/19...) — trials are not independent $\Rightarrow$ **not binomial** (it is hypergeometric, Unit 5).

(ii) Fixed $n = 50$, independent calls, constant $p = 0.7$, two outcomes $\Rightarrow$ **binomial** $B(50, 0.7)$.

WHERE MARKS ARE LOST

Writing $P(X \geq 2) = P(2) \cdot 0.9^6$ computed as 0.54 (keep 4+ decimals) · saying (i) is binomial “because each card is either defective or not” — two outcomes alone is not enough.

Q 7.2 Recover n and p from mean and variance · pmf · $E(X^2)$ · impossible range

5 marks

- A binomial random variable has mean 6 and variance 4.5. Find n and p , and write the pmf. [2]
- Find $E(X^2)$ and $P(X = 0)$. [1.5]
- Find $P(24 < X < 30)$. [0.5]
- A different binomial variable has $n = 30$ and mean 20. Find its variance and standard deviation. [1]

WHAT THIS QUESTION TESTS

Using $\text{Var}/E = q$ to recover the parameters, the identity $E(X^2) = \text{Var} + E^2$, and noticing that a binomial cannot exceed n .

HOW TO ATTEMPT

- $npq = 6$, $npq = 4.5 \Rightarrow q = 0.75$, $p = 0.25$, $n = 24$.
- $E(X^2) = 4.5 + 36$; $P(0) = q^n$.
- Values above 24 are impossible.
- $p = 20/30$, then npq .

ANSWERS AT A GLANCE

a) $q = 0.75$, $p = 0.25$, $n = 24$; $P(X = k) = \binom{24}{k} 0.25^k 0.75^{24-k}$ b) 40.5 and 0.0010 c) 0 d) Var 6.667, SD 2.582

STEP-BY-STEP SOLUTION**STEP** **part a** Divide variance by mean

$$\frac{npq}{np} = q = \frac{4.5}{6} = 0.75 \Rightarrow p = 0.25$$

$$n = \frac{6}{0.25} = 24$$

$$X \sim B(24, 0.25); P(X = k) = \binom{24}{k} (0.25)^k (0.75)^{24-k}, k = 0, 1, \dots, 24$$

STEP **part b** Second moment and $P(0)$

$$E(X^2) = \text{Var}(X) + [E(X)]^2 = 4.5 + 36 = 40.5$$

$$P(X = 0) = 0.75^{24} = 0.0010$$

STEP **part c** Impossible range

X can only take values 0 to 24, so no value satisfies $24 < X < 30$

$$P(24 < X < 30) = 0$$

STEP **part d** Variance from n and mean

$$p = 20/30 = 2/3, q = 1/3$$

$$\text{Var} = npq = 30 \times \frac{2}{3} \times \frac{1}{3} = 6.667; \text{SD} = 2.582$$

$$\text{Var } 6.667, \text{SD } 2.582$$

WHERE MARKS ARE LOST

Taking $p = 4.5/6$ (that is q) · $E(X^2) = E(X)^2$ · attempting to compute (c) with the formula instead of noticing the range.

Q 7.3 Binomial · MCQ guessing · at least / at most · mean and variance

4 marks

A quiz has 6 multiple-choice questions, each with 4 options and exactly one correct answer. A student guesses every answer at random. Let X be the number of correct answers.

- Probability of at least 4 correct answers. [1.5]
- Probability of at most 1 correct answer. [1.5]
- Mean and variance of X . [1]

WHAT THIS QUESTION TESTS

Binomial with a fraction p , where exact fractions (powers of 4) keep the arithmetic clean.

HOW TO ATTEMPT

1. $X \sim B(6, \frac{1}{4})$; work with denominator $4^6 = 4096$.
2. At least 4 = $P(4) + P(5) + P(6)$; at most 1 = $P(0) + P(1)$.

ANSWERS AT A GLANCE

a) $154/4096 = 0.0376$ b) $2187/4096 = 0.5339$ c) 1.5 and 1.125

STEP-BY-STEP SOLUTION**STEP** part a **At least 4**

$$P(4) = \binom{6}{4} \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^2 = 15 \cdot \frac{9}{4096} = \frac{135}{4096}$$

$$P(5) = \binom{6}{5} \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right) = 6 \cdot \frac{3}{4096} = \frac{18}{4096}$$

$$P(6) = \left(\frac{1}{4}\right)^6 = \frac{1}{4096}$$

$$P(X \geq 4) = \frac{135 + 18 + 1}{4096} = \frac{154}{4096} = 0.0376$$

0.0376**STEP** part b **At most 1**

$$P(0) = \left(\frac{3}{4}\right)^6 = \frac{729}{4096}; P(1) = 6 \cdot \frac{1}{4} \cdot \left(\frac{3}{4}\right)^5 = 6 \cdot \frac{243}{4096} = \frac{1458}{4096}$$

$$P(X \leq 1) = \frac{729 + 1458}{4096} = 0.5339$$

0.5339**STEP** part c **Mean and variance**

$$E(X) = 6 \times \frac{1}{4} = 1.5; \text{Var}(X) = 6 \times \frac{1}{4} \times \frac{3}{4} = 1.125$$

WHERE MARKS ARE LOST

Using $p = 0.5$ ("right or wrong") instead of $1/4$ · "at least 4" including only $P(4)$ · dropping the binomial coefficients.

Q 7.4 Poisson · rate scaling · at least · none · consecutive intervals**5 marks**

A help-line receives calls at an average rate of 2 per hour, following a Poisson process.

- a) Probability of exactly 3 calls in an hour. [1]
- b) Probability of at least 2 calls in a 3-hour period. [1.5]
- c) Probability of no calls in a 15-minute interval. [1]
- d) Probability of exactly one call in each of two consecutive 30-minute intervals. [1]
- e) State the mean and variance of the number of calls in a 3-hour period. [0.5]

WHAT THIS QUESTION TESTS

Rescaling λ to the interval in the question, the complement for "at least", and independence of disjoint intervals.

HOW TO ATTEMPT

1. Write λ for each interval: 1 h $\rightarrow$ 2; 3 h $\rightarrow$ 6; 15 min $\rightarrow$ 0.5; 30 min $\rightarrow$ 1.
2. Poisson formula; tabulate terms for the complement.
3. Disjoint intervals are independent $\Rightarrow$ multiply.

ANSWERS AT A GLANCE

a) 0.1804 b) 0.9826 c) 0.6065 d) 0.1353 e) both 6

STEP-BY-STEP SOLUTION**STEP** part a One hour, $\lambda = 2$

$$P(X = 3) = \frac{e^{-2}2^3}{3!} = \frac{0.1353 \times 8}{6} = 0.1804$$

STEP part b Three hours, $\lambda = 6$

$$P(X \geq 2) = 1 - P(0) - P(1) = 1 - e^{-6}(1 + 6) = 1 - 0.002479 \times 7 = 1 - 0.01735 = 0.9826$$

0.9826**STEP** part c 15 minutes, $\lambda = 0.5$

$$P(X = 0) = e^{-0.5} = 0.6065$$

STEP part d Two independent 30-minute intervals, $\lambda = 1$ each

$$P(X = 1) = e^{-1} \cdot 1 = 0.3679; \text{ both: } 0.3679^2 = 0.1353$$

0.1353 (note this is not the same as "exactly 2 calls in an hour", which is $e^{-2}2^2/2 = 0.2707$)**STEP** part e Mean = variance

For Poisson(6): mean 6, variance 6 (SD 2.449)

WHERE MARKS ARE LOST

Using $\lambda = 2$ for every part · "at least 2" as $1 - P(1)$ · in (d) computing $P(2 \text{ calls in 1 h})$ instead of the product.

Q 7.5 Poisson approximation to Binomial · expected counts · compare with exact**5 marks**

A pharmacy packs tablets in strips of 500. Each tablet is chipped with probability 0.004, independently. The pharmacy ships 2000 strips.

- a) Justify a Poisson approximation and state its parameter. [1]
- b) Probability that a strip has no chipped tablet; exactly one; at least three. [2]
- c) How many of the 2000 strips are expected to have no chipped tablet? [1]
- d) Compute the exact binomial probability of no chipped tablet and compare. [1]

WHAT THIS QUESTION TESTS

The large- n , small- p approximation, cumulative Poisson terms, and expected counts across many items.

HOW TO ATTEMPT

1. $n = 500$ large, $p = 0.004$ small $\Rightarrow \lambda = np = 2$.
2. Tabulate $P(0), P(1), P(2)$; at least 3 = 1 – their sum.
3. Expected count = $2000 \times P(0)$.
4. Exact: 0.996^{500} .

ANSWERS AT A GLANCEa) $\lambda = 2$ b) 0.1353, 0.2707, 0.3233 c) ≈ 271 strips d) Exact 0.1348 — agrees to 3 decimals**STEP-BY-STEP SOLUTION**

STEP **part a** Why Poisson

$X \sim B(500, 0.004)$; n is large and p is small with $np = 2$ moderate $\Rightarrow X \approx \text{Poisson}(\lambda = 2)$

STEP **part b** Poisson terms

k	$P(X = k) = e^{-2} 2^k / k!$
0	$e^{-2} = 0.1353$
1	$0.1353 \times 2 = 0.2707$
2	$0.2707 \times 2/2 = 0.2707$

$$P(X \geq 3) = 1 - (0.1353 + 0.2707 + 0.2707) = 1 - 0.6767 = 0.3233$$

$$P(0) = 0.1353, P(1) = 0.2707, P(X \geq 3) = 0.3233$$

STEP **part c** Expected count

$$2000 \times 0.1353 = 270.7$$

About 271 strips

STEP **part d** Exact binomial

$$P(X = 0) = 0.996^{500} = e^{500 \ln 0.996} = e^{-2.004} = 0.1348$$

0.1348 vs 0.1353 — the approximation is excellent here because p is very small.

WHERE MARKS ARE LOST

Using $\lambda = 0.004$ (forgetting to multiply by n) · "at least 3" as $1 - P(3)$ · reporting 270.7 strips instead of a whole number.

Q 7.6 Binomial vs Poisson · same probability both ways · explain the gap

3 marks

A biased coin lands heads with probability 0.10. It is tossed 30 times.

- a) Exact binomial probability of exactly 2 heads. [1]
- b) The same probability by the Poisson approximation. [1]
- c) Compare the two and explain when the approximation works well. [1]

WHAT THIS QUESTION TESTS

Doing the same calculation twice and explaining the difference in terms of the approximation's conditions.

HOW TO ATTEMPT

- $\binom{30}{2} 0.1^2 0.9^{28}$.
- $\lambda = 3$; $e^{-3} 3^2 / 2$.
- Absolute difference; comment on p not being tiny.

ANSWERS AT A GLANCE

a) 0.2277 b) 0.2240 c) difference 0.0037 — good but not perfect because $p = 0.1$ is only moderately small

STEP-BY-STEP SOLUTION**STEP** **part a** Exact

$$P(X = 2) = \binom{30}{2} (0.1)^2 (0.9)^{28} = 435 \times 0.01 \times 0.05233 = 0.2277$$

STEP **part b** Poisson, $\lambda = np = 3$

$$P(X = 2) = \frac{e^{-3} 3^2}{2!} = \frac{0.04979 \times 9}{2} = 0.2240$$

STEP **part c** Compare

Difference = $|0.2277 - 0.2240| = 0.0037$ (about 1.6 % relative error)

The Poisson slightly under-estimates here. The approximation is best when n is large and p is small (say $p < 0.05$, $n \geq 50$); with $p = 0.1$ and $n = 30$ it is usable but not exact, so the exact binomial should be preferred when it is computable.

WHERE MARKS ARE LOST

0.9^{28} rounded to 0.05 (gives 0.2175) · using $\lambda = 0.1$ · saying “Poisson is always more accurate for rare events” without reference to n and p .

Q 7.7 Normal · below, between, above · expected count

5 marks

The weight X (grams) of a packaged snack is normally distributed with mean 50 g and standard deviation 8 g.

- a) $P(X < 38)$. [1]
- b) $P(42 < X < 62)$. [2]
- c) $P(X > 66)$, expressed as a percentage. [1]
- d) In a batch of 400 packets, how many are expected to weigh more than 66 g? [1]

WHAT THIS QUESTION TESTS

The three basic normal shapes (left tail, middle, right tail), table reading with negative z , and turning a probability into a count.

HOW TO ATTEMPT

1. Convert each limit: $z = (x - 50)/8$.
2. Sketch which area is wanted; use Φ .
3. Count = $400 \times$ probability.

ANSWERS AT A GLANCE

a) $\Phi(-1.5) = 0.0668$ b) $\Phi(1.5) - \Phi(-1) = 0.9332 - 0.1587 = 0.7745$ c) $1 - \Phi(2) = 0.0228 = 2.28\%$ d) ≈ 9 packets

STEP-BY-STEP SOLUTION**STEP** **part a** Left tail

$$z = \frac{38 - 50}{8} = -1.5; P(X < 38) = \Phi(-1.5) = 1 - \Phi(1.5) = 1 - 0.9332 = 0.0668$$

0.0668

STEP **part b** Between

$$z_1 = \frac{42 - 50}{8} = -1, z_2 = \frac{62 - 50}{8} = 1.5$$

$$P = \Phi(1.5) - \Phi(-1) = 0.9332 - 0.1587 = 0.7745$$

0.7745

STEP **part c** Right tail

$$z = \frac{66 - 50}{8} = 2; P(X > 66) = 1 - \Phi(2) = 1 - 0.9772 = 0.0228$$

2.28 %**STEP** **part d** Expected count

$$400 \times 0.0228 = 9.1$$

About 9 packets**WHERE MARKS ARE LOST**

Reading $\Phi(-1.5)$ as 0.9332 (forgetting the symmetry flip) · subtracting the z -values instead of the areas · leaving (c) as 0.0228 when a percentage is asked.

Q 7.8 Inverse normal · cut-offs · verify a claim**5 marks**

Scores on a certification exam are modelled as $N(70, 10^2)$. The examining body makes three statements:

- a) "Most candidates score below 85." Find $P(X < 85)$ and say whether "most" is justified. [1]
- b) "The top 5 % receive a distinction." Find the distinction cut-off score. [1.5]
- c) "The bottom 10 % must re-sit." Find the re-sit cut-off. [1.5]
- d) Find the scores between which the middle 50 % of candidates lie. [1]

WHAT THIS QUESTION TESTS

Reading the z -table backwards (area $\rightarrow z$), then converting z back to x ; the quartiles of a normal.

HOW TO ATTEMPT

- (a) forward: $z = 1.5$.
- Top 5 % $\Rightarrow$ area to the left 0.95 $\Rightarrow z = 1.645$; $x = \mu + z\sigma$.
- Bottom 10 % $\Rightarrow$ area 0.10 $\Rightarrow z = -1.28$.
- Middle 50 % $\Rightarrow z = \pm 0.674$.

ANSWERS AT A GLANCE

a) 0.9332 — yes, 93 % is "most" b) 86.45 c) 57.2 d) 63.26 to 76.74

STEP-BY-STEP SOLUTION**STEP** **part a** Forward

$$z = (85 - 70)/10 = 1.5; P(X < 85) = \Phi(1.5) = 0.9332$$

93.3 % score below 85 — "most" is justified.**STEP** **part b** Top 5 %

$$\text{Need } P(X > c) = 0.05 \Rightarrow \Phi(z) = 0.95 \Rightarrow z = 1.645$$

$$c = 70 + 1.645 \times 10 = 86.45$$

Distinction cut-off ≈ 86.45 (a candidate needs 87 or more if scores are whole numbers)**STEP** **part c** Bottom 10 %

$$\Phi(z) = 0.10 \Rightarrow z = -1.28$$

$$c = 70 - 1.28 \times 10 = 57.2$$

Re-sit cut-off ≈ 57.2

STEP **part d** **Middle 50 %**

$$\Phi(z) = 0.25 \Rightarrow z = -0.674; \Phi(z) = 0.75 \Rightarrow z = 0.674$$

$$70 \pm 0.674 \times 10 = 63.26 \text{ and } 76.74$$

The middle half lies between 63.3 and 76.7 (these are the quartiles; IQR = 13.5 = 1.35σ)

WHERE MARKS ARE LOST

Using $z = 1.96$ for the top 5 % (1.96 is for 2.5 % in each tail) · forgetting the minus sign for the bottom 10 % · reporting z instead of the score.

Q 7.9 Specification limits · proportion defective · expected count · process comment**5 marks**

Bolts are produced with diameter $X \sim N(10.00, 0.05^2)$ mm. Specifications require 9.92 mm to 10.08 mm; bolts outside this range are rejected. A batch of 1000 bolts is produced.

- a) Proportion of bolts rejected. [2.5]
- b) Expected number rejected in the batch, and expected number with diameter above 10.10 mm. [1.5]
- c) Comment on the process capability. What would reduce the rejection rate: shifting the mean, or reducing σ ? [1]

WHAT THIS QUESTION TESTS

Two-sided specification limits (middle area, then its complement), counts, and a numbers-based comment.

HOW TO ATTEMPT

1. z for both limits: ± 1.6 .
2. Within = $\Phi(1.6) - \Phi(-1.6)$; rejected = 1 – within.
3. Counts = 1000 × probability.
4. Mean is already centred $\Rightarrow$ only reducing σ helps.

ANSWERS AT A GLANCE

a) 0.1096 (about 11 %) b) ≈ 110 rejected; ≈ 23 above 10.10 mm c) Not capable; reduce σ (mean is already at the target)

STEP-BY-STEP SOLUTION**STEP** **part a** **Two-sided**

$$z_L = \frac{9.92 - 10}{0.05} = -1.6, z_U = \frac{10.08 - 10}{0.05} = 1.6$$

$$P(\text{within}) = \Phi(1.6) - \Phi(-1.6) = 0.9452 - 0.0548 = 0.8904$$

$$P(\text{rejected}) = 1 - 0.8904 = 0.1096$$

10.96 % rejected

STEP **part b** **Counts**

$$\text{Rejected: } 1000 \times 0.1096 = 109.6 \approx 110$$

$$\text{Above 10.10: } z = 0.10/0.05 = 2; 1 - \Phi(2) = 0.0228; 1000 \times 0.0228 = 22.8 \approx 23$$

≈ 110 rejected; ≈ 23 bolts above 10.10 mm

STEP part c **Capability**

An 11 % rejection rate is too high for an automated process. The mean (10.00) already sits exactly at the centre of the specification, so shifting it would only make one tail worse. The spec half-width (0.08) is just 1.6σ ; the fix is to reduce σ (e.g. to 0.025 mm, making the limits $\pm 3.2\sigma$ and the rejection rate about 0.14 %).

WHERE MARKS ARE LOST

Computing only one tail and forgetting the other · $\Phi(-1.6)$ read as 0.9452 · commenting "the process is fine" without comparing the rate to any standard.

Q 7.10 Normal approximation to Binomial · continuity correction

5 marks

10 % of viewers skip an advertisement within the first 5 seconds. A random sample of 400 viewers is taken. Let X be the number who skip.

- a) Verify that the normal approximation may be used and state μ and σ . [1]
- b) Estimate $P(30 \leq X \leq 50)$. [2]
- c) Estimate $P(X > 45)$. [1.5]
- d) Why is the continuity correction needed? [0.5]

WHAT THIS QUESTION TESTS

Checking $np, nq \geq 15$, computing μ, σ , and applying ± 0.5 correctly for " $\leq / \geq$ " and for ">".

HOW TO ATTEMPT

- $np = 40, nq = 360$ both ≥ 15 ✓.
- $30 \leq X \leq 50 \rightarrow 29.5 < X < 50.5$.
- $X > 45 \rightarrow X > 45.5$.

ANSWERS AT A GLANCE

a) $\mu = 40, \sigma = 6$ b) $\Phi(1.75) - \Phi(-1.75) = 0.9599 - 0.0401 = 0.9198$ c) $1 - \Phi(0.92) = 1 - 0.8212 = 0.1788$ d) a discrete count is being approximated by a continuous curve

STEP-BY-STEP SOLUTION**STEP** part a **Conditions and parameters**

$$np = 400(0.1) = 40 \geq 15, nq = 400(0.9) = 360 \geq 15 \checkmark$$

$$\mu = np = 40; \sigma = \sqrt{npq} = \sqrt{36} = 6$$

$$X \approx N(40, 6^2)$$

STEP part b **Between, inclusive**

$$\text{Continuity: } P(30 \leq X \leq 50) \approx P(29.5 < X < 50.5)$$

$$z_1 = \frac{29.5 - 40}{6} = -1.75, z_2 = \frac{50.5 - 40}{6} = 1.75$$

$$P = \Phi(1.75) - \Phi(-1.75) = 0.9599 - 0.0401 = 0.9198$$

0.9198

STEP **part c** More than 45

$$P(X > 45) \approx P(X > 45.5); z = \frac{45.5 - 40}{6} = 0.92$$

$$P = 1 - \Phi(0.92) = 1 - 0.8212 = 0.1788$$

0.1788**STEP** **part d** Why ± 0.5

The binomial is a set of bars of width 1 centred on integers; the normal is a smooth curve. Extending each integer's bar half a unit on either side makes the curve area match the bar area — otherwise integer end-points are half-counted.

WHERE MARKS ARE LOST

Using 30 and 50 directly (gives 0.9050, loses a mark) · correcting $X > 45$ to 44.5 instead of 45.5 · $\sigma = npq = 36$ without the square root.

Q 7.11 Recover parameters · μ and σ from two probabilities · λ from $P(0)$ **4 marks**

- a) Delivery times are normally distributed. 20 % of deliveries take less than 60 minutes and 90 % take less than 80 minutes. Find μ and σ . [3]
- b) The number of faults per day on a network follows a Poisson distribution, and 5 % of days are fault-free. Find the average number of faults per day and $P(X = 2)$. [1]

WHAT THIS QUESTION TESTS

Working the normal table backwards twice and solving two simultaneous equations; recovering λ from a single Poisson probability.

HOW TO ATTEMPT

- $\Phi(z_1) = 0.20 \Rightarrow z_1 = -0.84$; $\Phi(z_2) = 0.90 \Rightarrow z_2 = 1.28$.
- Equations $\mu - 0.84\sigma = 60$, $\mu + 1.28\sigma = 80$; subtract.
- $e^{-\lambda} = 0.05 \Rightarrow \lambda = -\ln 0.05$.

ANSWERS AT A GLANCE

a) $\sigma = 9.43$, $\mu = 67.9$ min b) $\lambda = 3.0$; $P(X = 2) = 0.224$

STEP-BY-STEP SOLUTION**STEP** **part a** Two equations

$$P(X < 60) = 0.20 \Rightarrow \frac{60 - \mu}{\sigma} = -0.84 \Rightarrow \mu - 0.84\sigma = 60$$

$$P(X < 80) = 0.90 \Rightarrow \frac{80 - \mu}{\sigma} = 1.28 \Rightarrow \mu + 1.28\sigma = 80$$

$$\text{Subtract: } 2.12\sigma = 20 \Rightarrow \sigma = 9.43$$

$$\mu = 60 + 0.84 \times 9.43 = 67.92$$

 $\mu \approx 67.9$ min, $\sigma \approx 9.4$ min**STEP** **part b** λ from the zero probability

$$P(X = 0) = e^{-\lambda} = 0.05 \Rightarrow \lambda = -\ln 0.05 = 3.0$$

$$P(X = 2) = \frac{e^{-3} 3^2}{2} = 0.2240$$

Average 3 faults per day; $P(X = 2) = 0.224$

WHERE MARKS ARE LOST

Using $z = +0.84$ for the 20 % point · adding the equations instead of subtracting · $\lambda = \ln 0.05$ (negative).

UNIT 8

Sampling Distributions & Estimation

Population vs sample · parameter vs statistic · sampling distribution of the mean · Central Limit Theorem · standard error · finite population correction · sample proportion · confidence interval for μ (σ known) · sample size for a margin of error

What this unit is about

A sample mean is itself a random quantity — take another sample and you get another mean. The **sampling distribution of $\bar{X}$** describes how those means vary. Its centre is the population mean μ ; its spread, the **standard error** $\sigma/\sqrt{n}$, shrinks as the sample grows. The **Central Limit Theorem** says its shape is (approximately) normal once $n \geq 30$, whatever the population looks like — and exactly normal for any n if the population is normal. That single fact lets you use the z-table for questions about sample means and sample proportions exactly as in Unit 7, with σ replaced by the standard error.

Estimation runs the idea backwards: from one sample, a **confidence interval** $\bar{x} \pm z_{\alpha/2} \sigma/\sqrt{n}$ gives a range that captures μ with the stated confidence; the half-width is the **margin of error**, and solving the margin-of-error formula for n tells you how big a sample a survey needs.

How to attempt questions from this unit

1. **Identify what is random:** a single value (X , use σ), a sample mean ($\bar{X}$, use $\sigma/\sqrt{n}$), or a sample proportion ($\hat{p}$, use $\sqrt{p(1-p)/n}$). Writing the standard error explicitly is usually worth a mark.
2. **Convert to z** with the standard error in the denominator, then use the table as in Unit 7.
3. **Finite population:** if $n/N > 0.05$, multiply the standard error by $\sqrt{(N-n)/(N-1)}$.
4. **Proportion:** check $np \geq 15$ and $n(1-p) \geq 15$ before using the normal.
5. **Confidence interval:** choose $z_{\alpha/2}$ (1.645 for 90 %, 1.96 for 95 %, 2.576 for 99 %), compute margin $E = z\sigma/\sqrt{n}$, interval $\bar{x} \pm E$. Interpret as "we are 95 % confident the population mean lies between ... and ...".
6. **Sample size:** $n = \left(\frac{z\sigma}{E}\right)^2$, always round up.

Formula card

Mean of $\bar{X}$: $\mu_{\bar{X}} = \mu$

CLT: $\bar{X} \approx N(\mu, \sigma^2/n)$ for $n \geq 30$ (any n if population normal)

Finite population: $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$ (use if $n/N > 0.05$)

$$z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$$

Margin of error $E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$z_{\alpha/2}$: 90 % $\rightarrow$ 1.645, 95 % $\rightarrow$ 1.96, 98 % $\rightarrow$ 2.33, 99 % $\rightarrow$ 2.576

Standard error: $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Proportion: $\hat{p} \approx N\left(p, \frac{p(1-p)}{n}\right)$, needs $np, nq \geq 15$

CI for μ (σ known): $\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

Sample size $n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2$ (round up)

Halving E needs 4 $\times$ the sample

Question map

Q	Shape of the question	Marks
8.1	Probability for a sample mean: between two values, above a value; effect of a larger sample	4
8.2	Sample mean from a finite population: with and without the correction factor	4
8.3	Sample proportion: forecasting a winner, probability of a wrong call	4
8.4	Confidence intervals at 95 % and 99 %, margin of error, interpretation	4
8.5	Sample size for a required margin of error at two confidence levels; halving the margin	4
8.6	Objective checks on the CLT, standard error, and a “does a binomial fit?” consistency test	4

Q 8.1 Sampling distribution of the mean · CLT · effect of n

4 marks

The time a customer spends at a self-checkout has mean 40 s and standard deviation 6 s (distribution unknown). A random sample of 36 customers is observed.

- a) State the approximate distribution of the sample mean $\bar{X}$, with justification. [1]
- b) Probability that the sample mean lies between 38 and 42 s. [1.5]
- c) Probability that the sample mean exceeds 41.5 s. [1]
- d) If the sample size were 144 instead, what happens to the standard error and to the probability in (b)? [0.5]

WHAT THIS QUESTION TESTS

Writing the CLT statement, the standard error, and z-scores for a sample mean; the $1/\sqrt{n}$ effect.

HOW TO ATTEMPT

- $n = 36 \geq 30 \Rightarrow \text{CLT}; \text{SE} = 6/\sqrt{36} = 1.$
- $z = (\bar{x} - 40)/1.$
- $n = 144 \Rightarrow \text{SE halves to } 0.5.$

ANSWERS AT A GLANCE

a) $\bar{X} \approx N(40, 1^2)$ b) 0.9544 c) 0.0668 d) SE = 0.5; probability ≈ 1 ($z = \pm 4$)

STEP-BY-STEP SOLUTION

STEP part a CLT

$n = 36 \geq 30$, so by the Central Limit Theorem $\bar{X} \approx N(\mu, \sigma^2/n) = N(40, 36/36) = N(40, 1)$; SE = $6/\sqrt{36} = 1$ s

STEP part b Between

$$z_1 = \frac{38 - 40}{1} = -2, z_2 = \frac{42 - 40}{1} = 2$$

$$P = \Phi(2) - \Phi(-2) = 0.9772 - 0.0228 = 0.9544$$

0.9544

STEP part c Above

$$z = \frac{41.5 - 40}{1} = 1.5; P(\bar{X} > 41.5) = 1 - \Phi(1.5) = 1 - 0.9332 = 0.0668$$

0.0668

STEP **part d** Larger sample

$$SE = 6/\sqrt{144} = 0.5 \text{ (halved)}; z = \pm 2/0.5 = \pm 4; P \approx \Phi(4) - \Phi(-4) \approx 1$$

Quadrupling n halves the SE; the sample mean is almost certain to lie within ± 2 s of 40.

WHERE MARKS ARE LOST

Using $\sigma = 6$ instead of $\sigma/\sqrt{n} = 1$ (gives 0.2586 for (b) — the single most common error) · not stating the CLT · claiming SE is divided by 4 when n is multiplied by 4.

Q 8.2 Finite population correction · sample mean**4 marks**

A factory employs 500 workers whose monthly overtime has mean 18 h and standard deviation 6 h. A random sample of 80 workers is taken.

- a)** Should the finite population correction be used? Compute the standard error of the sample mean with it. [1.5]
- b)** Probability that the sample mean overtime exceeds 19.5 h. [1.5]
- c)** Recompute (b) without the correction and comment. [1]

WHAT THIS QUESTION TESTS

The $n/N > 0.05$ rule, the correction factor, and its effect on a tail probability.

HOW TO ATTEMPT

1. $n/N = 80/500 = 0.16 > 0.05 \Rightarrow$ use FPC.
2. $SE = \frac{6}{\sqrt{80}} \sqrt{\frac{420}{499}}.$
3. $z = (19.5 - 18)/SE.$

ANSWERS AT A GLANCE

a) Yes; $SE = 0.6708 \times 0.9174 = 0.6154$ b) $z = 2.44, P = 0.0073$ c) $z = 2.24, P = 0.0125$ — ignoring the correction overstates the probability

STEP-BY-STEP SOLUTION**STEP** **part a** Check and compute SE

$$n/N = 80/500 = 0.16 > 0.05 \Rightarrow \text{apply the correction}$$

$$\frac{\sigma}{\sqrt{n}} = \frac{6}{\sqrt{80}} = \frac{6}{8.944} = 0.6708$$

$$\sqrt{\frac{N-n}{N-1}} = \sqrt{\frac{420}{499}} = \sqrt{0.8417} = 0.9174$$

$$SE = 0.6708 \times 0.9174 = 0.6154$$

SE = 0.6154 h

STEP **part b** Tail probability

$$z = \frac{19.5 - 18}{0.6154} = 2.44; P(\bar{X} > 19.5) = 1 - \Phi(2.44) = 1 - 0.9927 = 0.0073$$

0.0073

STEP **part c** Without correction

$$z = \frac{1.5}{0.6708} = 2.24; P = 1 - \Phi(2.24) = 1 - 0.9875 = 0.0125$$

0.0125 — the uncorrected SE is too large, so the tail probability is overstated (0.0125 vs 0.0073). Sampling 16 % of a finite population genuinely reduces the variability of the mean.

WHERE MARKS ARE LOST

Using $\sqrt{(N-n)/N}$ · applying the correction to σ before dividing by $\sqrt{n}$ and then dividing again · not checking the 5 % rule.

Q 8.3 Sampling distribution of a proportion · forecasting

4 marks

A candidate has the support of 55 % of all voters. An exit poll samples 200 voters and will declare the candidate the winner if the sample proportion is at least 0.60.

- a) Verify that the normal approximation applies, and state the mean and standard error of $\hat{p}$. [1.5]
- b) Probability that the poll declares the candidate the winner. [1.5]
- c) Probability that the sample proportion falls below 0.50 (a wrong call of defeat). [1]

WHAT THIS QUESTION TESTS

The proportion version of the CLT with its own standard error.

HOW TO ATTEMPT

1. $np = 110, nq = 90 \geq 15$.
2. $SE = \sqrt{0.55 \times 0.45 / 200}$.
3. $z = (\hat{p} - 0.55) / SE$ for 0.60 and 0.50 (symmetric).

ANSWERS AT A GLANCE

a) mean 0.55, SE 0.0352 b) $z = 1.42, P = 0.0778$ c) $z = -1.42, P = 0.0778$

STEP-BY-STEP SOLUTION**STEP** **part a** Conditions and SE

$$np = 200(0.55) = 110 \geq 15, nq = 90 \geq 15 \checkmark$$

$$\mu_{\hat{p}} = p = 0.55; SE = \sqrt{\frac{0.55 \times 0.45}{200}} = \sqrt{0.0012375} = 0.0352$$

STEP **part b** Declared winner

$$z = \frac{0.60 - 0.55}{0.0352} = 1.42; P(\hat{p} \geq 0.60) = 1 - \Phi(1.42) = 1 - 0.9222 = 0.0778$$

0.0778 — with a 0.60 threshold the poll will usually *not* call a true 55 % winner

STEP **part c** Wrong call of defeat

$$z = \frac{0.50 - 0.55}{0.0352} = -1.42; P(\hat{p} < 0.50) = \Phi(-1.42) = 0.0778$$

0.0778

WHERE MARKS ARE LOST

Using $\sqrt{p(1-p)}$ without dividing by n · dividing by n outside the square root · using $\hat{p}$ (0.60) instead of p (0.55) in the SE.

Q 8.4 Confidence interval for μ · margin of error · interpretation

4 marks

A random sample of 64 customer calls has mean handling time 52 s. From long records the population standard deviation is 8 s.

- a) Find the margin of error at 95 % confidence and the 95 % confidence interval for the mean handling time. [2]
- b) Find the 99 % confidence interval. [1]
- c) Interpret the 95 % interval, and explain why the 99 % interval is wider. [1]

WHAT THIS QUESTION TESTS

The CI formula with σ known, and the trade-off between confidence and width.

HOW TO ATTEMPT

1. $SE = 8/\sqrt{64} = 1$.
2. $E = 1.96 \times 1$; interval $52 \pm E$.
3. Replace 1.96 by 2.576.

ANSWERS AT A GLANCE

a) $E = 1.96$; (50.04, 53.96) b) (49.42, 54.58) c) 95 % confident μ is between 50.04 and 53.96 s; more confidence needs a larger $z \Rightarrow$ wider

STEP-BY-STEP SOLUTION**STEP** part a 95 % interval

$$SE = \frac{8}{\sqrt{64}} = 1; z_{0.025} = 1.96$$

$$E = 1.96 \times 1 = 1.96$$

$$\text{Interval: } 52 \pm 1.96 = (50.04, 53.96)$$

Margin 1.96 s; 95 % CI (50.04 s, 53.96 s)

STEP part b 99 % interval

$$z_{0.005} = 2.576; E = 2.576; 52 \pm 2.576 = (49.42, 54.58)$$

99 % CI (49.42 s, 54.58 s)

STEP part c Interpretation

We are 95 % confident that the true mean handling time of all calls lies between 50.04 s and 53.96 s (if we repeated the sampling many times, about 95 % of such intervals would contain μ). To be 99 % confident we must use $z = 2.576$ instead of 1.96, so the interval widens from 3.92 s to 5.15 s — more confidence costs precision.

WHERE MARKS ARE LOST

Using σ instead of $\sigma/\sqrt{n}$ · saying “there is a 95 % probability that μ is in the interval” (μ is fixed; the interval is random) · using 2.33 for 99 % (that is 98 %).

Q 8.5 Sample size for a margin of error · two confidence levels · halving E

4 marks

A canteen wants to estimate the mean time workers spend at lunch. A pilot study suggests $\sigma \approx 14$ minutes.

- a) How large a sample is needed to estimate the mean to within 2.5 minutes with 99 % confidence? [2]
 b) With 95 % confidence? [1]
 c) At 99 % confidence, how large a sample is needed to be within 1.25 minutes? Comment. [1]

WHAT THIS QUESTION TESTS

Rearranging the margin-of-error formula for n , rounding up, and the “halving E quadruples n ” rule.

HOW TO ATTEMPT

1. $n = (z\sigma/E)^2$ with $z = 2.576$.
2. Same with $z = 1.96$.
3. E halved $\Rightarrow n \times 4$.

ANSWERS AT A GLANCE

a) $n = 208.1 \rightarrow 209$ b) $n = 120.5 \rightarrow 121$ c) $n = 832.4 \rightarrow 833$ (four times (a))

STEP-BY-STEP SOLUTION**STEP** part a 99 %

$$n = \left(\frac{2.576 \times 14}{2.5} \right)^2 = \left(\frac{36.064}{2.5} \right)^2 = (14.426)^2 = 208.1$$

Round up: $n = 209$

STEP part b 95 %

$$n = \left(\frac{1.96 \times 14}{2.5} \right)^2 = (10.976)^2 = 120.5$$

$n = 121$

STEP part c Halve the margin

$$n = \left(\frac{2.576 \times 14}{1.25} \right)^2 = (28.852)^2 = 832.4$$

$n = 833$ — halving the margin of error quadruples the required sample ($n \propto 1/E^2$).

WHERE MARKS ARE LOST

Rounding 208.1 down to 208 (then the margin is slightly bigger than allowed) · forgetting to square · using the sample SD from a different question.

Q 8.6 Objective checks · CLT · standard error · binomial consistency

4 marks

- a) A population is normal with mean 100 and SD 15. For samples of size 9, state the exact distribution of $\bar{X}$. [1]
- b) A population is strongly right-skewed with mean 100 and SD 15. For samples of size 50, what can be said about the distribution of $\bar{X}$? Give its mean and standard error. [1]
- c) By what factor does the standard error change if the sample size goes from 25 to 100? [1]
- d) A sample of size 40 from a supposedly binomial population has sample mean 8 and sample SD 2. Is this consistent with $X \sim B(40, p)$? Support your answer. [1]

WHAT THIS QUESTION TESTS

Short conceptual items that appear as 1-mark parts: when normality is exact vs approximate, the $1/\sqrt{n}$ law, and checking a claimed distribution against its mean–variance relation.

HOW TO ATTEMPT

1. Normal population $\Rightarrow \bar{X}$ exactly normal for any n .
2. Skewed but $n \geq 30 \Rightarrow$ approximately normal (CLT).
3. $SE \propto 1/\sqrt{n}$.
4. Binomial: mean $np = 8 \Rightarrow p = 0.2$; implied SD $\sqrt{npq}$; compare with 2.

ANSWERS AT A GLANCE

a) $\bar{X} \sim N(100, 5^2)$ exactly b) approximately normal by CLT, mean 100, SE 2.12 c) halves d) implied SD $2.53 \neq 2 \Rightarrow$ not consistent

STEP-BY-STEP SOLUTION**STEP** **part a** Normal population

$$SE = 15/\sqrt{9} = 5$$

$\bar{X} \sim N(100, 25)$ exactly — no sample-size condition is needed when the population itself is normal.

STEP **part b** Skewed population, $n = 50$

$$SE = 15/\sqrt{50} = 2.12$$

By the CLT ($n = 50 \geq 30$), $\bar{X}$ is approximately $N(100, 2.12^2)$ even though individual values are skewed.

STEP **part c** Factor

$$\frac{\sigma/\sqrt{100}}{\sigma/\sqrt{25}} = \frac{5}{10} = \frac{1}{2}$$

The SE halves.

STEP **part d** Consistency with a binomial

$$\text{If } X \sim B(40, p): np = 8 \Rightarrow p = 0.2$$

$$\text{Implied SD} = \sqrt{40 \times 0.2 \times 0.8} = \sqrt{6.4} = 2.53$$

The observed SD is 2, not 2.53, so the mean and SD do not satisfy the binomial relation $\text{Var} = np(1 - p)$; the data is not consistent with $B(40, p)$.

WHERE MARKS ARE LOST

Saying (a) needs $n \geq 30$ · dividing the SE by 4 in (c) · in (d) stopping at $p = 0.2$ without checking the SD.

Cheat Sheet — everything on two pages

1 · Descriptive statistics

Sort first. Positions: $Q1 \frac{n+1}{4}$, Med $\frac{n+1}{2}$, $Q3 \frac{3(n+1)}{4}$; interpolate fractional positions.

$IQR = Q3 - Q1$ · fences $Q1 - 1.5 IQR$, $Q3 + 1.5 IQR$ · major: $3 IQR$ · on the fence ≠ outlier.

$$\bar{x} = \sum x/n \cdot s^2 = \sum (x - \bar{x})^2 / (n - 1) \cdot \sigma^2 \text{ divides by } N.$$

CV = $s/\bar{x} \times 100\%$ (lower = more consistent).

Mean > Median ⇒ right skew · Mean < Median ⇒ left skew · Mode ≈ 3 Med - 2 Mean.

Median for skewed/outlier data; mean otherwise.

Whiskers end at the last value inside the fences.

2 · Probability rules

$$P(A^c) = 1 - P(A) \cdot P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Exactly one: $P(A) + P(B) - 2P(A \cap B)$ · Neither: $1 - P(A \cup B)$ · Only A: $P(A) - P(A \cap B)$.

Mutually exclusive: $P(A \cap B) = 0$. Independent: $P(A \cap B) = P(A)P(B)$, $P(A | B) = P(A)$. Positive-probability events cannot be both.

Three events: $P(\cup) = \sum P(A_i) - \sum P(A_i A_j) + P(ABC)$ · exactly two = $\sum \text{pairs} - 3P(ABC)$ · only A = $P(A) - P(AB) - P(AC) + P(ABC)$.

Counting: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ · at least one = $1 - \text{none}$ · without replacement: shrinking denominators.

3 · Conditional · total probability · Bayes

$$P(B | A) = \frac{P(A \cap B)}{P(A)} \cdot P(A \cap B) = P(A)P(B | A) \cdot$$

chain: $P(A)P(B | A)P(C | AB)$.

Total: $P(E) = \sum_i P(C_i)P(E | C_i)$.

Bayes: $P(C_k | E) = \frac{P(C_k)P(E | C_k)}{P(E)}$; posteriors sum to 1.

Sensitivity = $P(+ | D)$, specificity = $P(- | D^c)$, false-positive rate = $1 - \text{specificity}$. Rare condition ⇒ low PPV.

Draw the tree; multiply along paths; add paths for the effect; one path ÷ total for a posterior.

4 · Naïve Bayes

Score(C) = $P(C) \prod_i P(x_i | C)$; pick the largest; posterior = score ÷ $\sum$ scores.

$P(x_i | C)$ = count of x_i within class C ÷ class size (use only the values in the new record).

Zero count ⇒ product 0 ⇒ Laplace: categorical $\frac{n_{x,C} + 1}{n_C + k}$; text $\frac{\text{count}(w, C) + 1}{N_C + |V|}$, $|V|$ = distinct training words.

Repeated word in a message ⇒ multiply its likelihood that many times.

5 · Discrete random variables

$\sum p(x) = 1$ finds k (reject negative roots) · $F(x) = P(X \leq x)$.

$$E(X) = \sum xp \cdot E(X^2) = \sum x^2 p \cdot \text{Var} = E(X^2) - [E(X)]^2.$$

$$E(aX + b) = aE(X) + b \cdot \text{Var}(aX + b) = a^2 \text{Var}(X).$$

Joint table: margins = row/col sums · conditional = cells ÷ the margin conditioned on · independent ⇔ every cell = row × col.

$$E(XY) = \sum \sum xy f(x, y) \cdot \text{Cov} = E(XY) - E(X)E(Y)$$

.

Without replacement: $P(X = k) = \binom{K}{k} \binom{N-K}{n-k} / \binom{N}{n}$, $E = nK/N$. Fair game ⇔ $E(\text{gain}) = 0$.

Reading the question — word → symbol

"of those who A, x % B" → $P(B | A) = x$ · "x % are both" → $P(A \cap B)$ · "or / at least one" → union · "neither" → $(A \cup B)^c$.

"given / if / knowing that" → the event after it goes in the denominator.

"at least k" → $1 - P(X \leq k - 1)$ · "at most k" → $P(X \leq k)$ · "more than k" → $1 - P(X \leq k)$ · "fewer than k" → $P(X \leq k - 1)$ · "between a and b inclusive" → $P(a) + \dots + P(b)$.

"per hour ... in 3 hours" → rescale λ · "sample of n ... mean of the sample" → use SE · "to within E" → margin of error.

Cheat Sheet — page 2

6 · Continuous random variables

$\int f = 1$ finds k (one integral per piece) · $P(a < X < b) = \int_a^b f = F(b) - F(a)$ · $P(X = a) = 0$.

$F(x) = \int_{-\infty}^x f$ · $f = F'$ · constant in a CDF from $F(\text{top}) = 1$ · median: $F(m) = 0.5$.

$E(X) = \int xf$ · $E(X^2) = \int x^2 f$ · $\int_0^a x^n = \frac{a^{n+1}}{n+1}$.

Joint: $f_X(x) = \int f dy$, $f_Y(y) = \int f dx$ · product on a rectangle $\Rightarrow$ independent; a sum never factors.

If independent: $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$, $E(XY) = E(X)E(Y)$.

7 · Standard distributions

Bernoulli(p): $E = p$, $\text{Var} = p(1 - p)$.

Binomial: $\binom{n}{k} p^k q^{n-k}$, $E = np$, $\text{Var} = npq$; from E , Var : $q = \text{Var}/E$, $n = E/p$. Needs fixed n , independent trials, constant p .

Poisson: $e^{-\lambda} \lambda^k / k!$, $E = \text{Var} = \lambda$; rescale λ to the interval; $P(k+1) = P(k)\lambda/(k+1)$; $\lambda = -\ln P(0)$.

Binomial $\rightarrow$ Poisson: n large, p small, $\lambda = np$.

Binomial $\rightarrow$ Normal: $np, nq \geq 15$, $\mu = np$, $\sigma = \sqrt{npq}$, continuity ± 0.5 : $\leq k \rightarrow < k + 0.5$, $\geq k \rightarrow > k - 0.5$, $< k \rightarrow < k - 0.5$, $> k \rightarrow > k + 0.5$.

Normal: $z = (x - \mu)/\sigma$ · $P(X < a) = \Phi(z)$ · $P(X > a) = 1 - \Phi(z)$ · $\Phi(-z) = 1 - \Phi(z)$ · cut-off: $x = \mu + z\sigma$.

68 – 95 – 99.7 rule · quartiles $\mu \pm 0.674\sigma$ · expected count = $N \times \text{probability}$.

Intro only: χ^2 (sum of squared normals, df k , right-skewed, 0 to ∞); t (normal $\div \sqrt{(\chi^2/\text{df})}$, heavier tails, small samples with unknown σ); F (ratio of two χ^2/df , compares variances).

8 · Sampling & estimation

$\bar{X} \approx N(\mu, \sigma^2/n)$ for $n \geq 30$ (exact for any n if the population is normal). $\text{SE} = \sigma/\sqrt{n}$.

Finite population ($n/N > 0.05$): multiply SE by $\sqrt{(N - n)/(N - 1)}$.

Proportion: $\hat{p} \approx N(p, p(1 - p)/n)$, needs $np, nq \geq 15$.

CI: $\bar{x} \pm z_{\alpha/2}\sigma/\sqrt{n}$ · $E = z\sigma/\sqrt{n}$ · $n = (z\sigma/E)^2$ round up · halve $E \Rightarrow 4 \times n$.

$z_{\alpha/2}$: 90 % 1.645 · 95 % 1.96 · 98 % 2.33 · 99 % 2.576.

"95 % confident μ lies in the interval" — never "95 % probability μ is in it".

Standard normal values you will use most · $\Phi(z) = P(Z \leq z)$

z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$
0.50	0.6915	1.28	0.8997	2.00	0.9772
0.674	0.7500	1.50	0.9332	2.24	0.9875
0.80	0.7881	1.60	0.9452	2.33	0.9901
0.84	0.7995	1.645	0.9500	2.44	0.9927
0.92	0.8212	1.75	0.9599	2.50	0.9938
1.00	0.8413	1.96	0.9750	2.576	0.9950
1.25	0.8944	1.42	0.9222	3.00	0.9987

For negative z use $\Phi(-z) = 1 - \Phi(z)$. Useful constants:

$e^{-1} = 0.3679$, $e^{-2} = 0.1353$, $e^{-3} = 0.0498$, $e^{-0.5} = 0.6065$, $e^{-6} = 0.00248$.

Exam-day checklist

1 · Define events / random variables in one line. 2 · Formula in symbols before numbers. 3 · One computation per line. 4 · 4 decimals until the end. 5 · Box the final answer with its unit. 6 · Interpretation = two sentences quoting your numbers. 7 · Check: probabilities in $[0,1]$, posteriors sum to 1, pmf sums to 1, area under a pdf = 1.